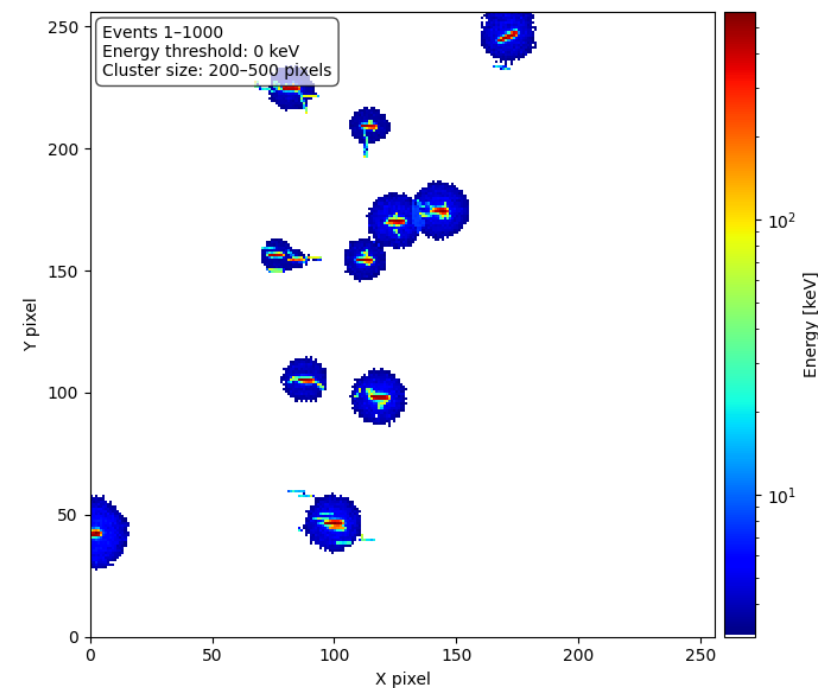
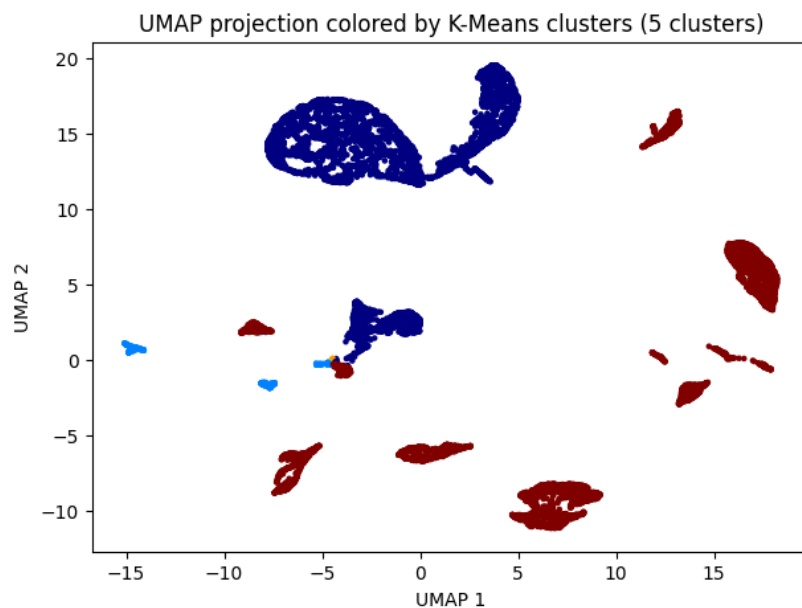
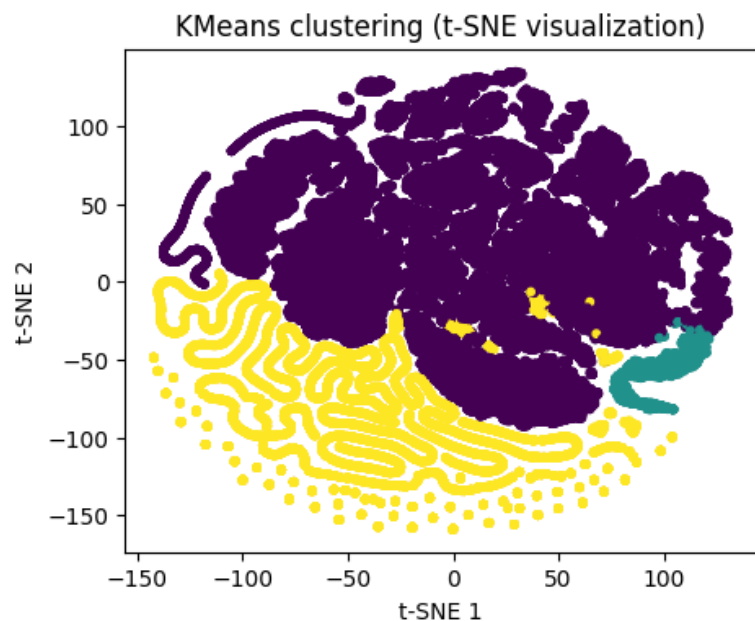
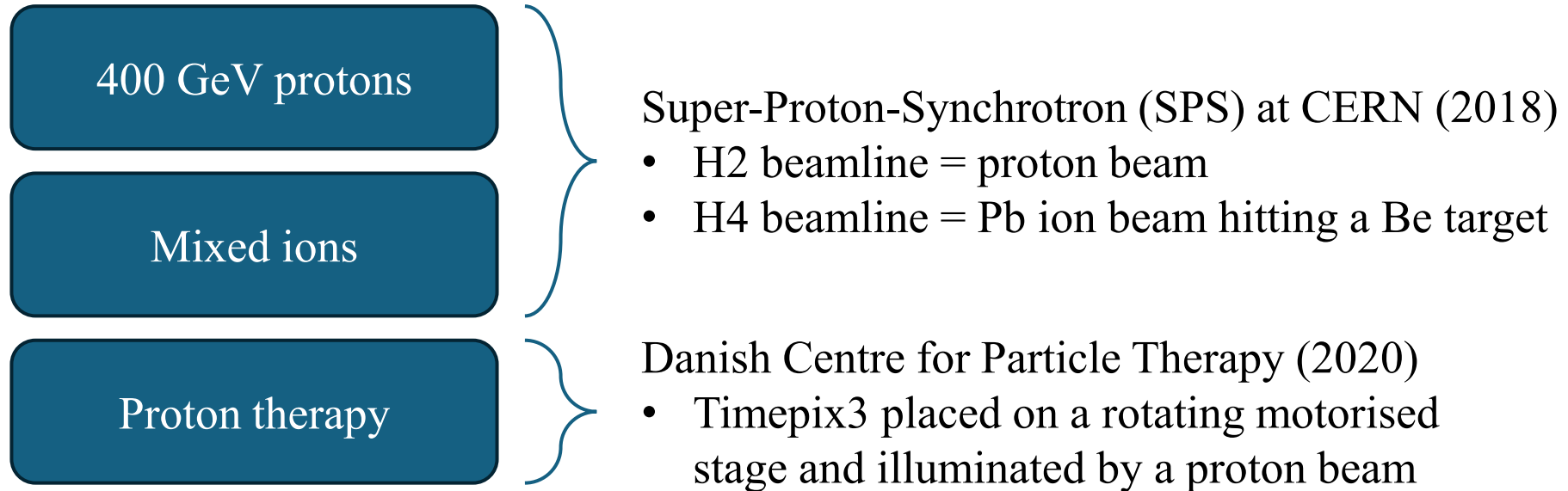


Unsupervised learning on Timepix3 data



Data preparation



- 1st step: converting root files into .dat files with five columns

X

Y

ToT

ToA

Event

- 2nd step: filtering out events on the edge (if X or Y == 0 or 255 the whole event is disregarded)

Feature Extraction

Extracted 13 features for each event, creating a (events, 13) dimensional matrix

Total energy

Maximum
energy

Mean energy

Standard
deviation of
energy

Number of hits

x spread

y spread

Width

Height

Aspect ratio

Inner pixels

Outer pixels

Volcano to total
pixel ratio

Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

PCA

t-SNE

UMAP

Clustering

K-means

Partitions data into k clusters by assigning points to the nearest centroid and iteratively updating those centroids.

`n_clusters`

DBSCAN

Groups together points that are densely packed (based on a radius and minimum neighbours) and marks low-density points as noise.

`eps`

`min_samples`

Gaussian Mixture

Assumes data is generated from a mix of Gaussian distributions and assigns probabilities of belonging to each cluster.

`n_components`

Bayesian Gaussian Mixture

Like GMM but uses Bayesian priors to automatically favour simpler models and prune unnecessary clusters.

`n_components`

`weight_concentration_prior`

Algorithm	Must choose k?	Cluster shape	Handles noise?	Hard/soft assignment	Speed
K-Means	Yes	Spherical blobs	No	Hard	Very fast
DBSCAN	No	Arbitrary	Yes	Hard (labels + noise)	Medium
GMM	Yes	Ellipses (Gaussian)	No	Soft (probabilities)	Medium
BGMM	Max only	Ellipses (Gaussian)	No	Soft	Slowest

Visualisation

PCA

(Principal Component Analysis): linearly reduces dimensionality by projecting data onto directions (components) that maximize variance.

n_components

t-SNE

(t-distributed Stochastic Neighbour Embedding): non-linear method that maps data into low dimensions while preserving local neighbourhood relationships.

n_components

perplexity

UMAP

(Uniform Manifold Approximation and Projection): non-linear dimensionality reduction that preserves both local and some global structure using graph-based learning.

n_neighbors

n_components

min_dist

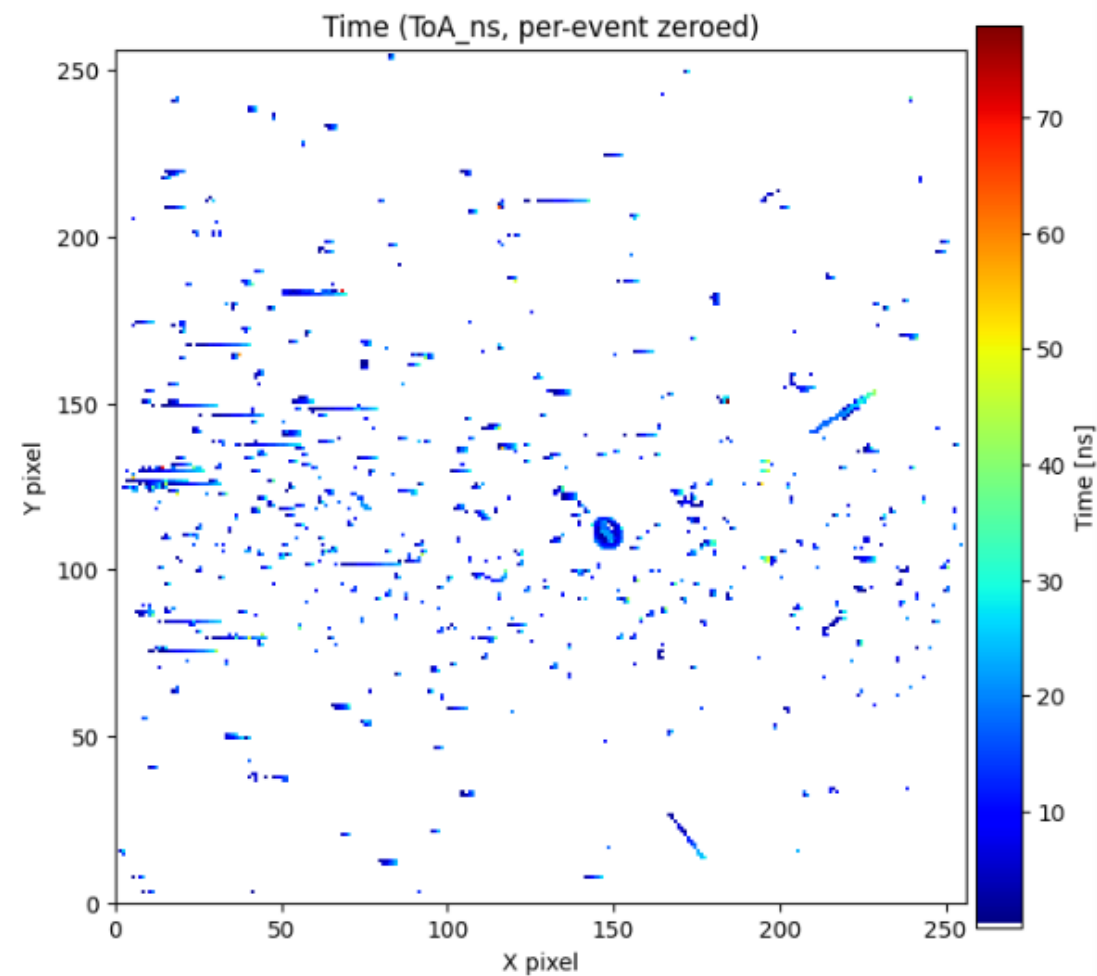
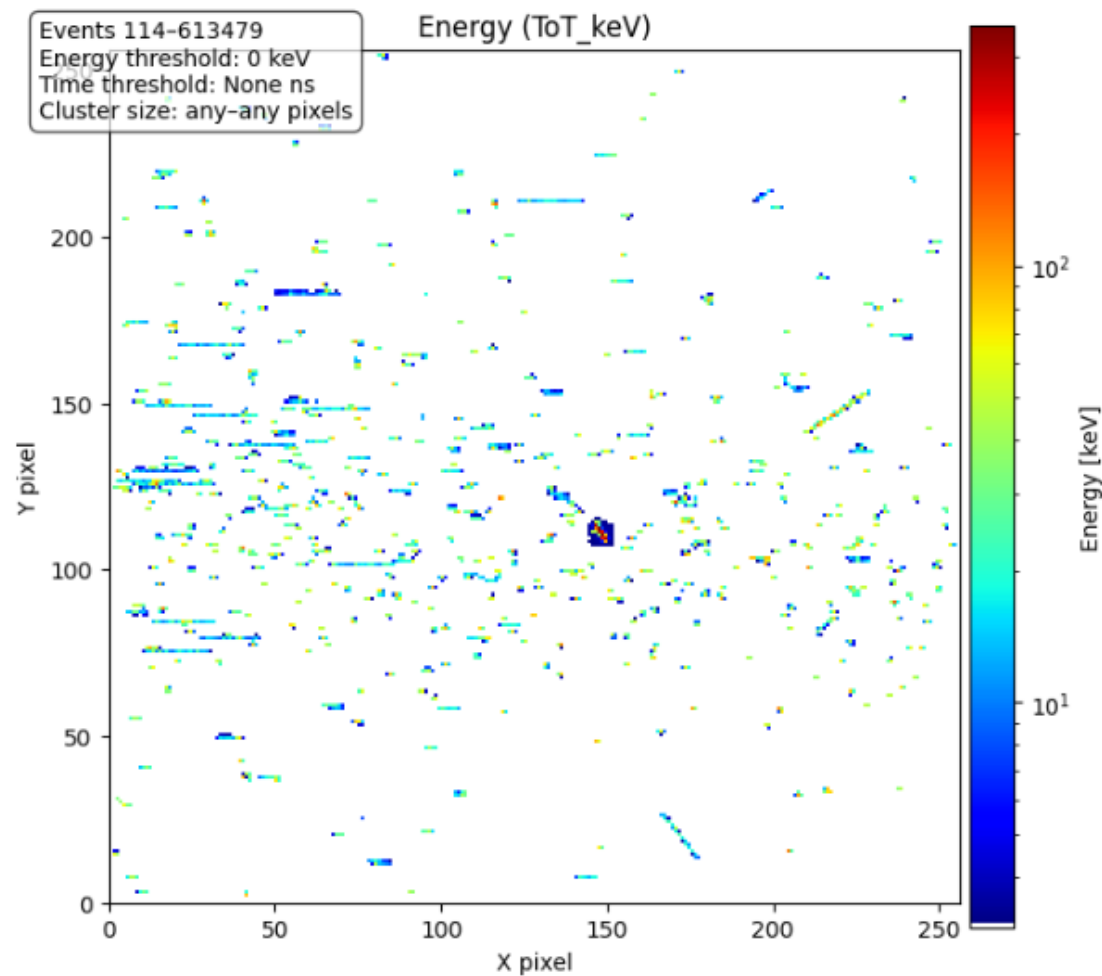
15deg

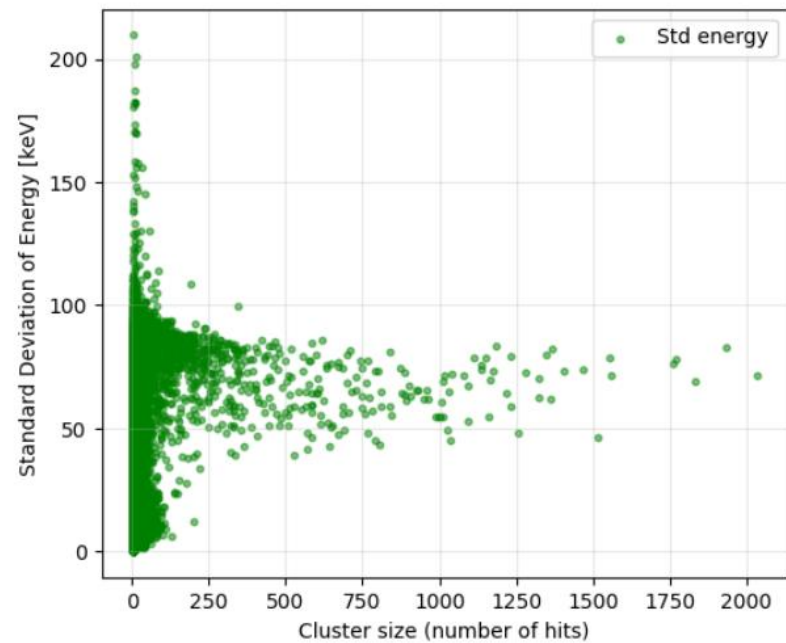
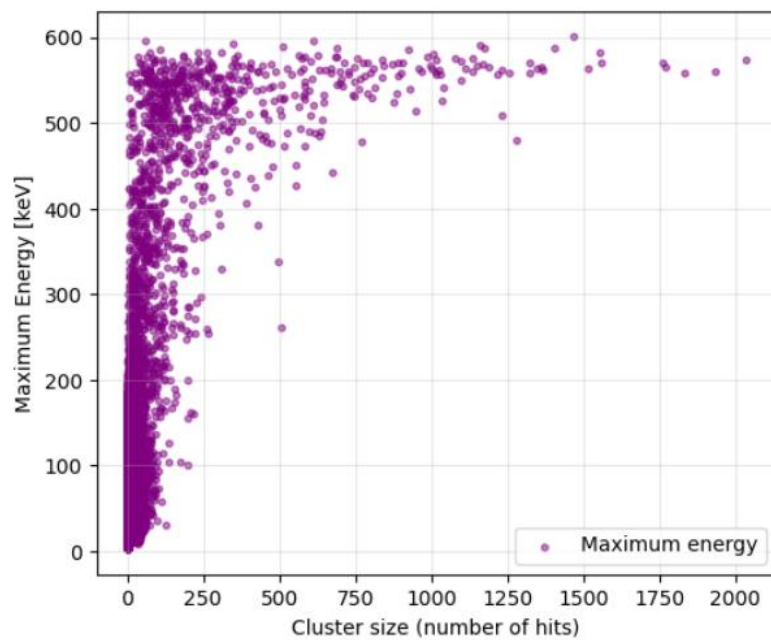
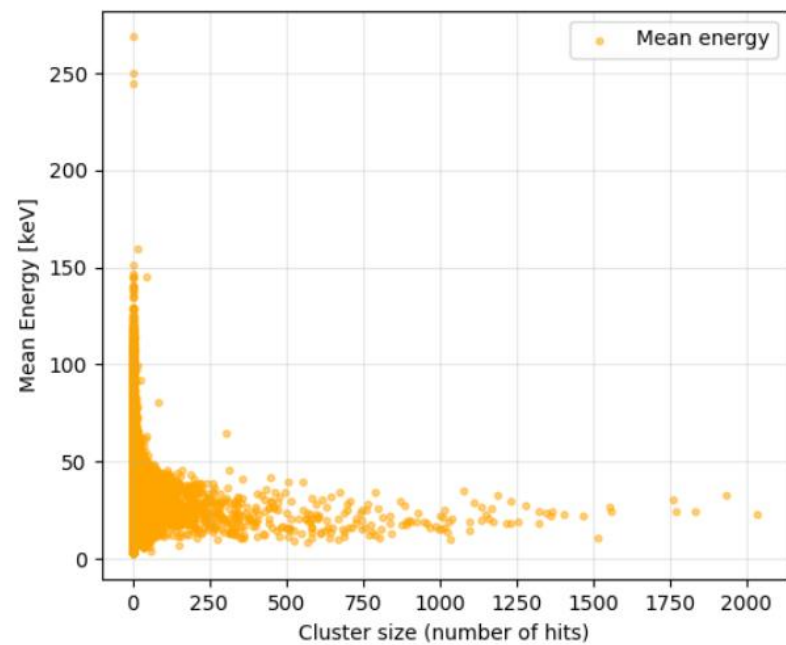
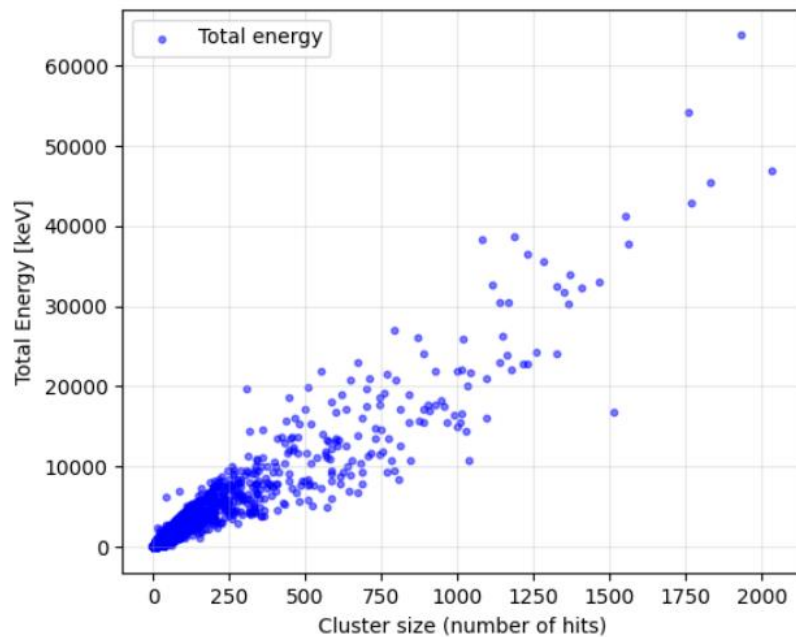
30deg

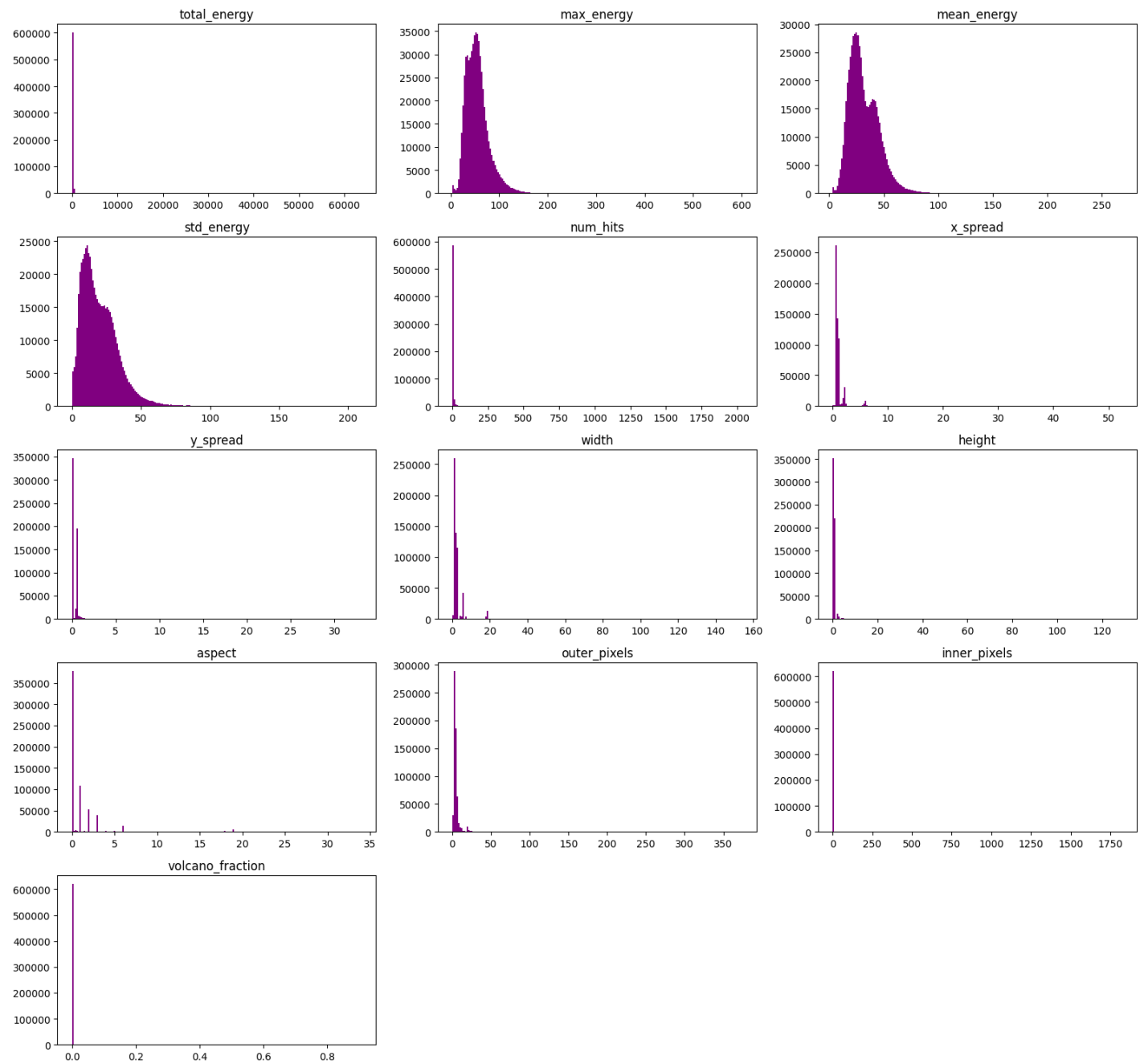
50deg

75deg

400 GeV protons







Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

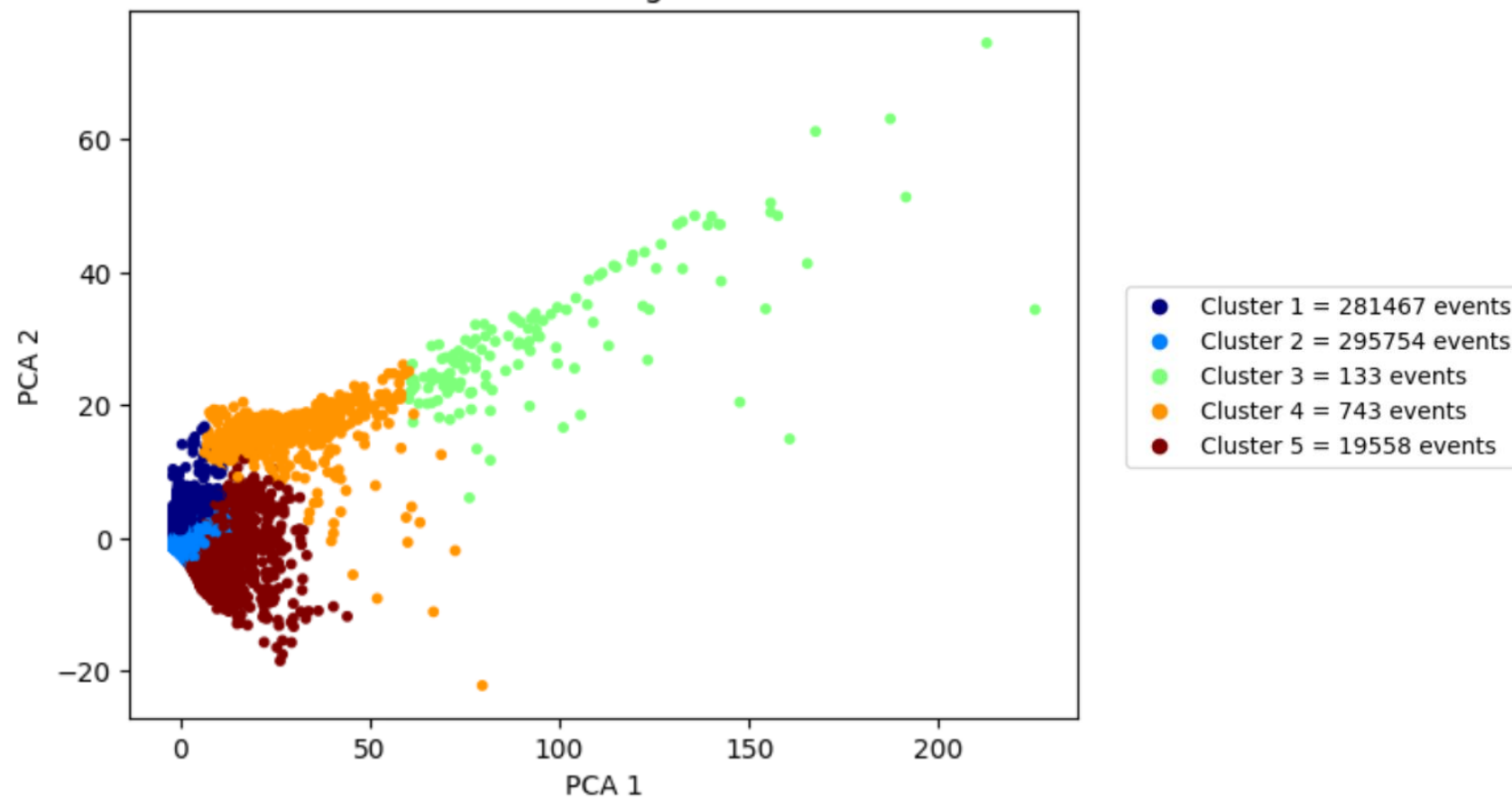
Visualisation

PCA

t-SNE

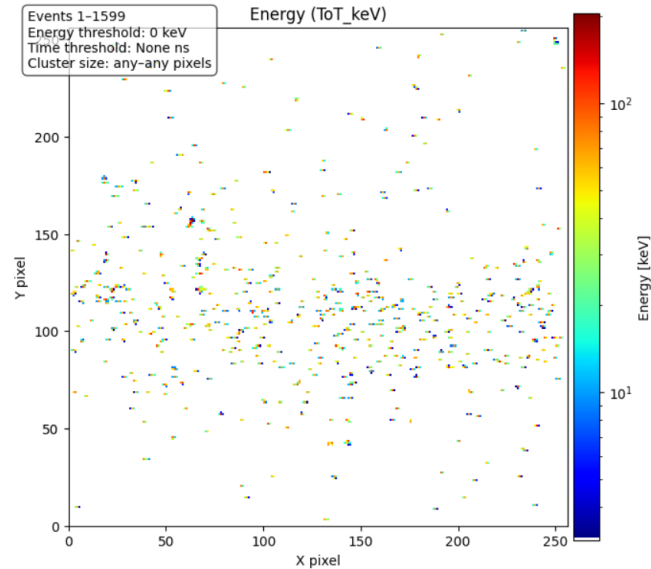
UMAP

K-means Clustering - 5 clusters

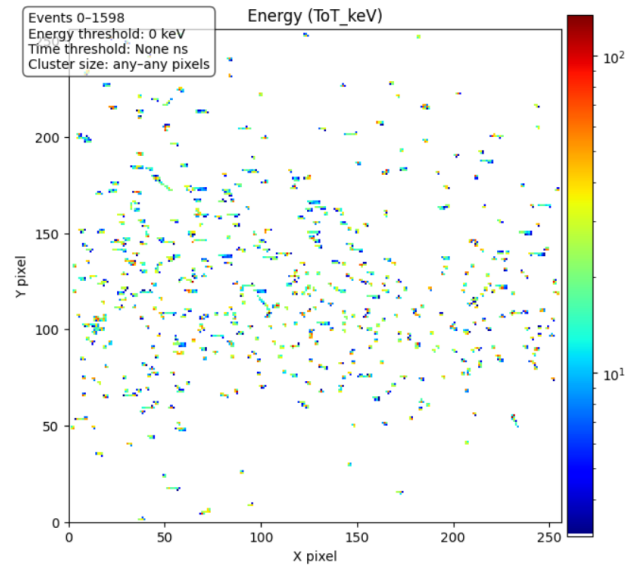


Cluster 1:

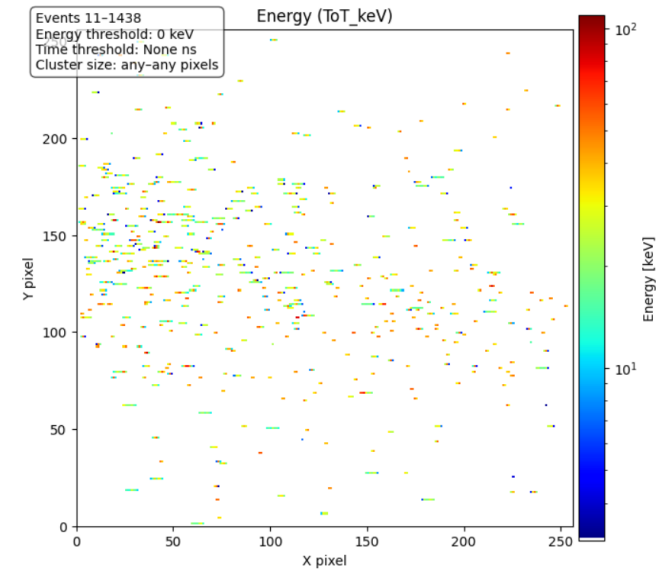
std_energy (cluster mean = 0.990, diff = 0.990)
 mean_energy (cluster mean = 0.882, diff = 0.882)
 max_energy (cluster mean = 0.781, diff = 0.781)
 Plotting 500 events from cluster 1...



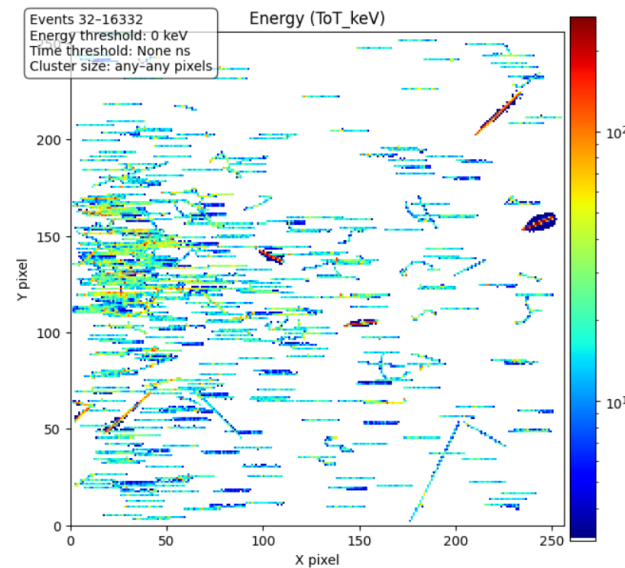
Cluster 2:
 y_spread (cluster mean = 0.783, diff = 0.783)
 mean_energy (cluster mean = -0.547, diff = 0.547)
 height (cluster mean = 0.494, diff = 0.494)
 Plotting 500 events from cluster 2...



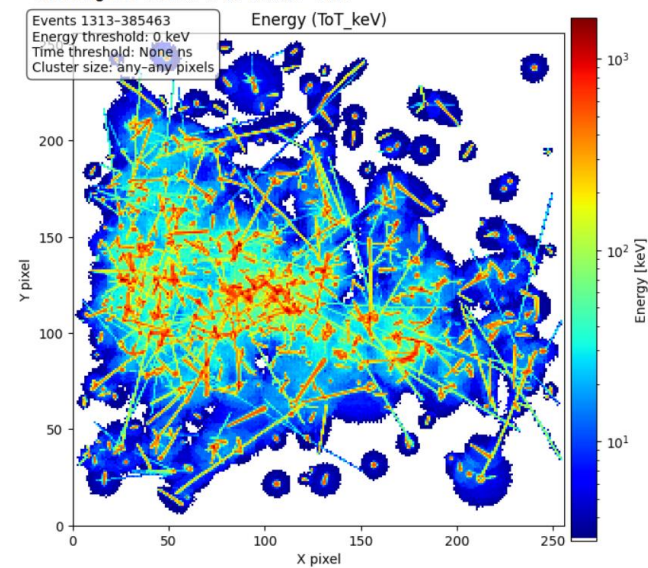
Cluster 3:
 std_energy (cluster mean = -0.728, diff = 0.728)
 max_energy (cluster mean = -0.562, diff = 0.562)
 y_spread (cluster mean = -0.540, diff = 0.540)
 Plotting 500 events from cluster 3...



Cluster 4:
 width (cluster mean = 4.735, diff = 4.735)
 x_spread (cluster mean = 4.705, diff = 4.705)
 outer_pixels (cluster mean = 3.777, diff = 3.777)
 Plotting 500 events from cluster 4...



Cluster 5:
 volcano_fraction (cluster mean = 24.481, diff = 24.481)
 total_energy (cluster mean = 17.150, diff = 17.150)
 num_hits (cluster mean = 17.017, diff = 17.017)
 Plotting 500 events from cluster 5...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

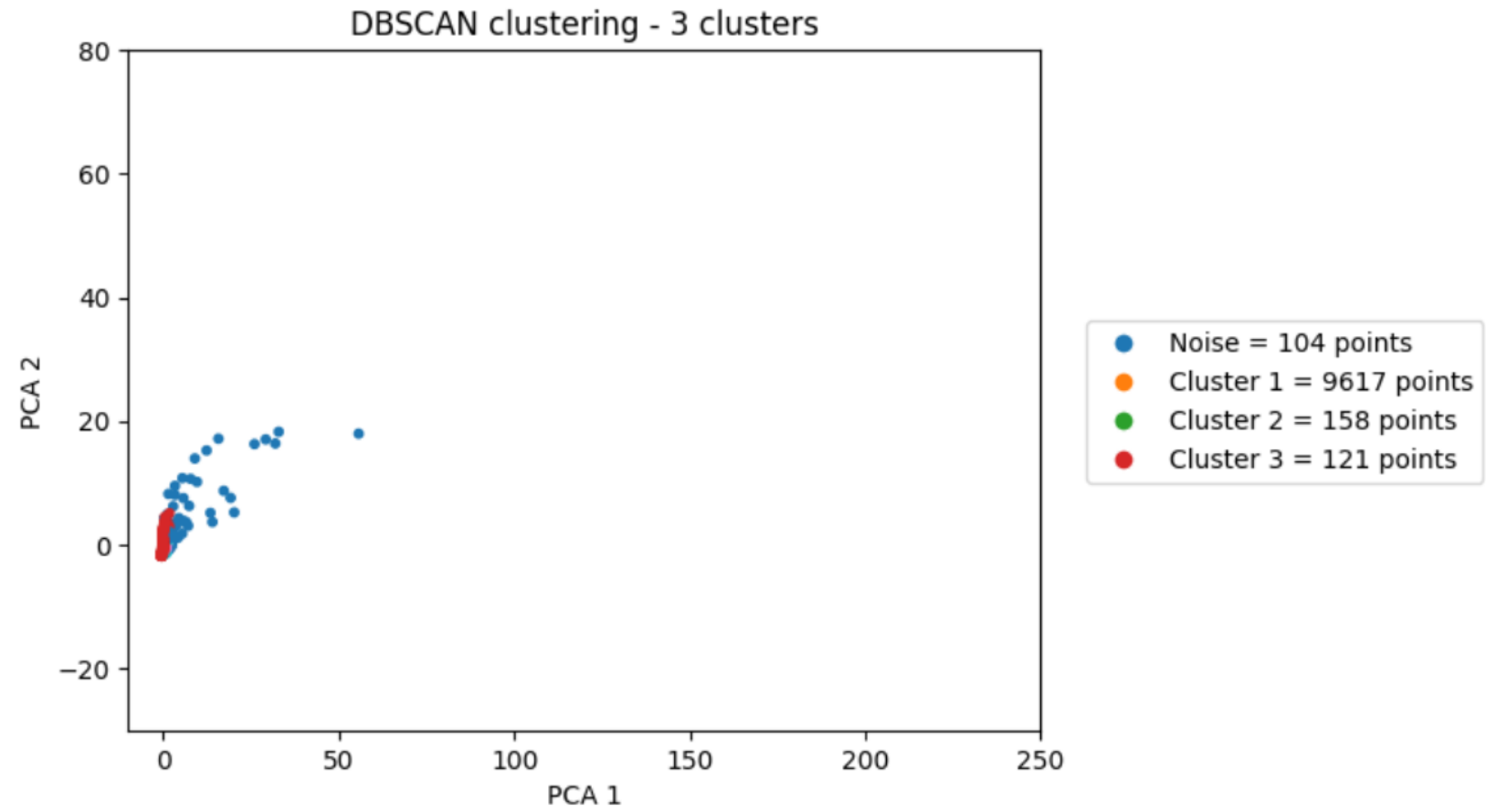
PCA

t-SNE

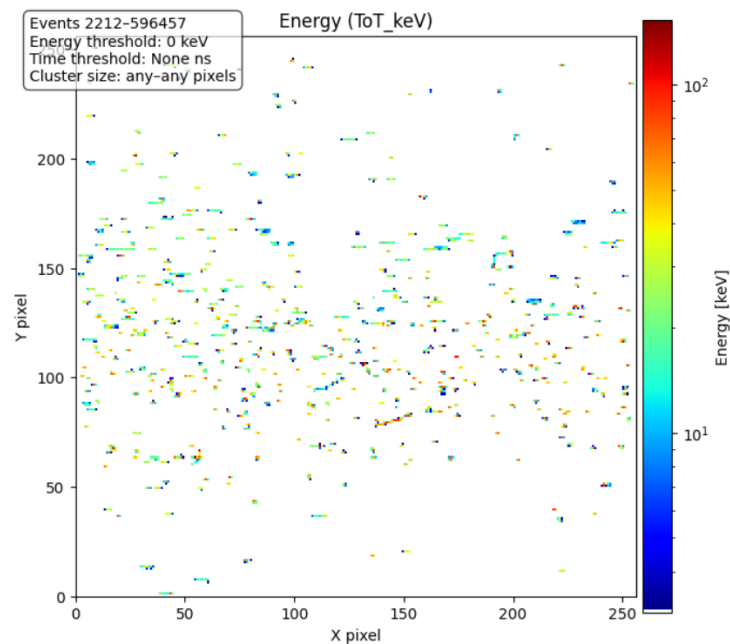
UMAP

DBSCAN eps (neighborhood radius): 2

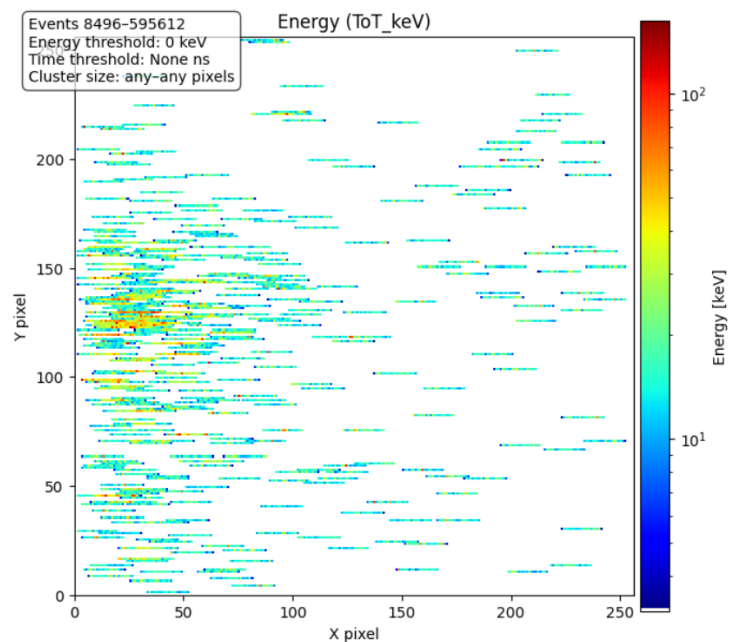
DBSCAN min_samples (minimum cluster size): 20



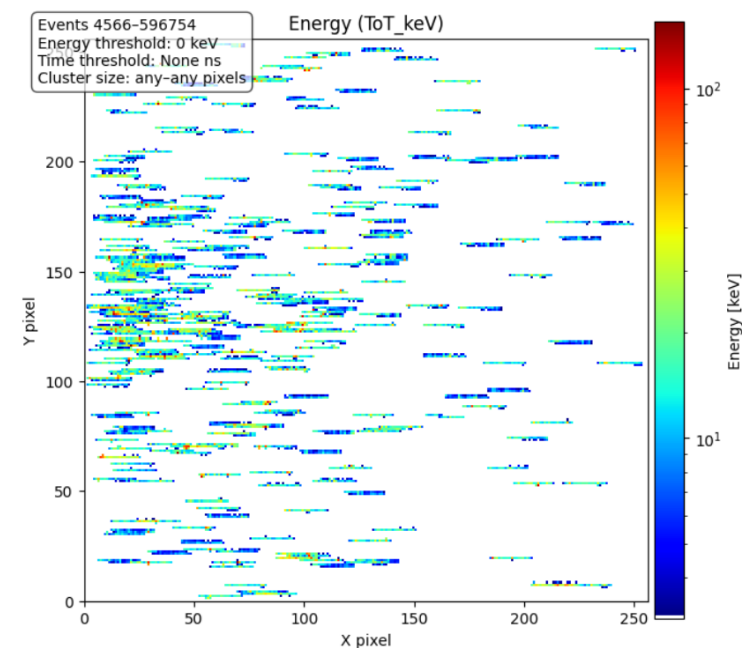
Cluster 1:
width (cluster mean = -0.155, diff = 0.155)
x_spread (cluster mean = -0.151, diff = 0.151)
outer_pixels (cluster mean = -0.136, diff = 0.136)
Plotting 500 events from cluster 1...



Cluster 2:
x_spread (cluster mean = 4.910, diff = 4.910)
width (cluster mean = 4.869, diff = 4.869)
outer_pixels (cluster mean = 2.990, diff = 2.990)
Plotting 422 events from cluster 2...



Cluster 3:
aspect (cluster mean = 7.784, diff = 7.784)
width (cluster mean = 4.864, diff = 4.864)
x_spread (cluster mean = 4.736, diff = 4.736)
Plotting 323 events from cluster 3...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

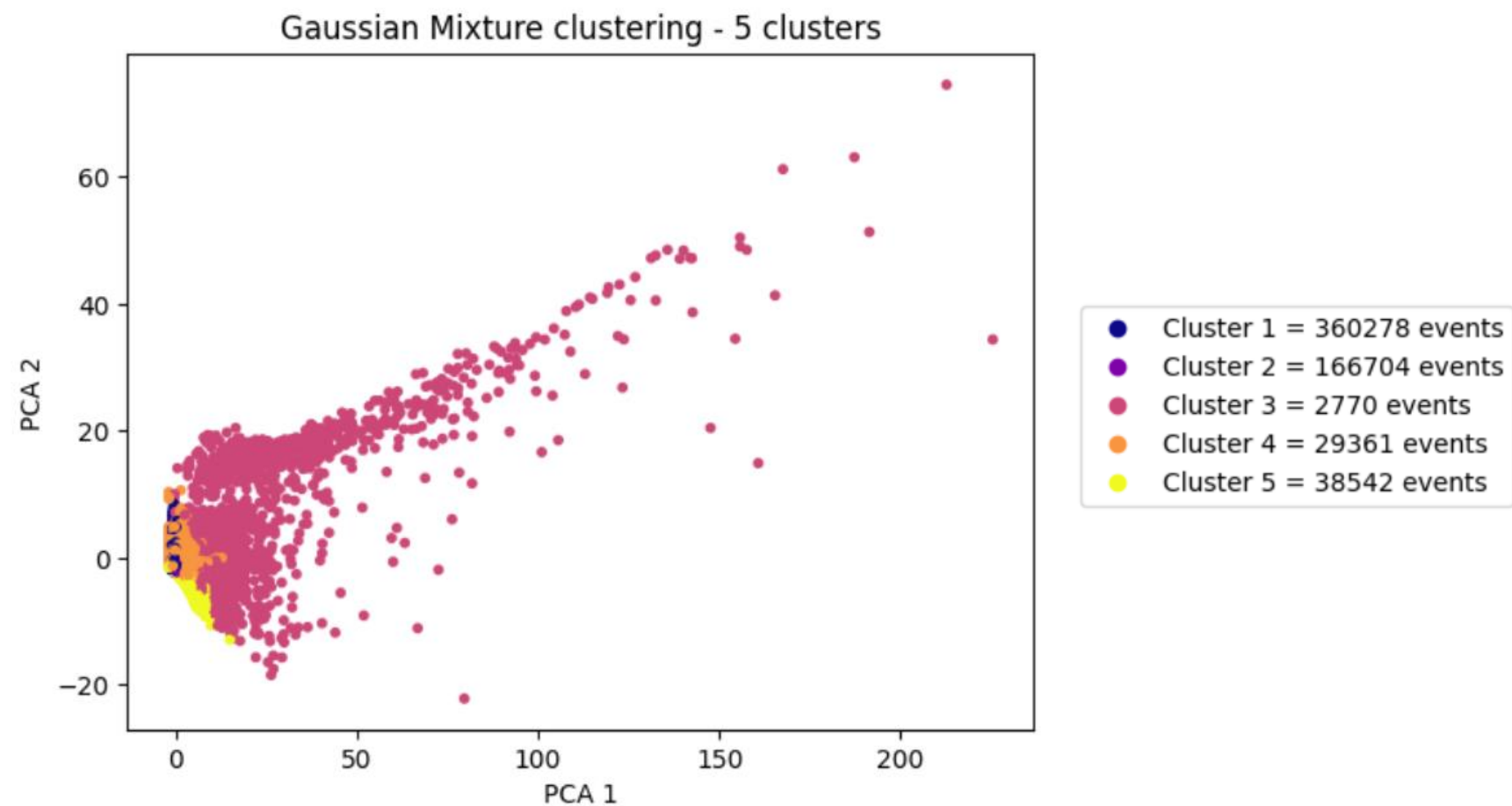
Visualisation

PCA

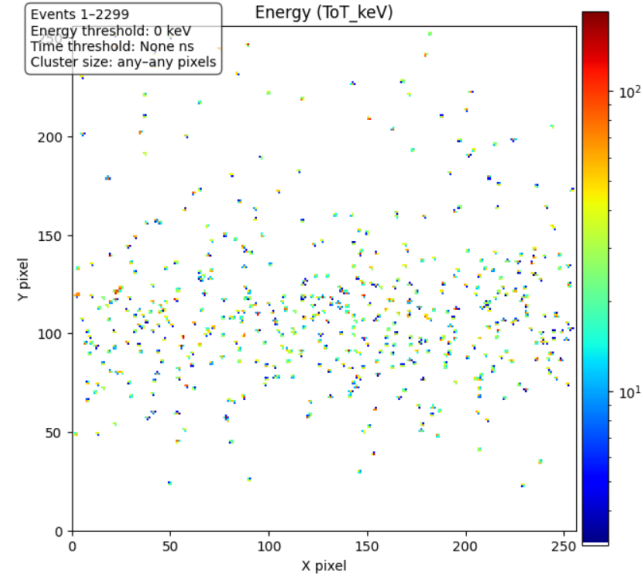
t-SNE

UMAP

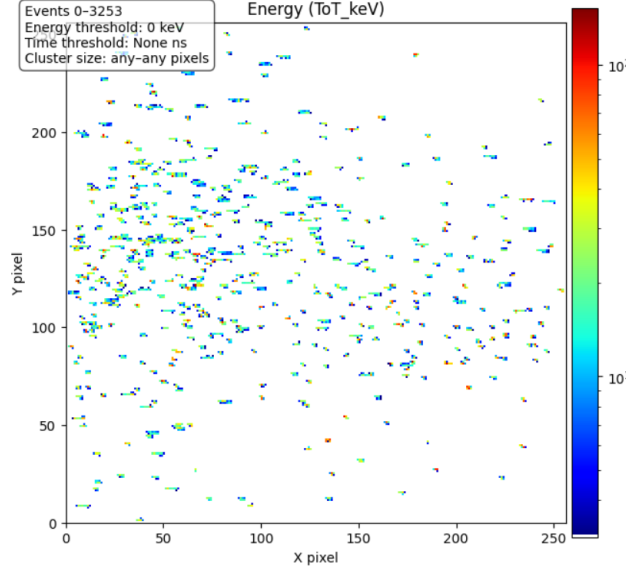
Number of Gaussian clusters: 5



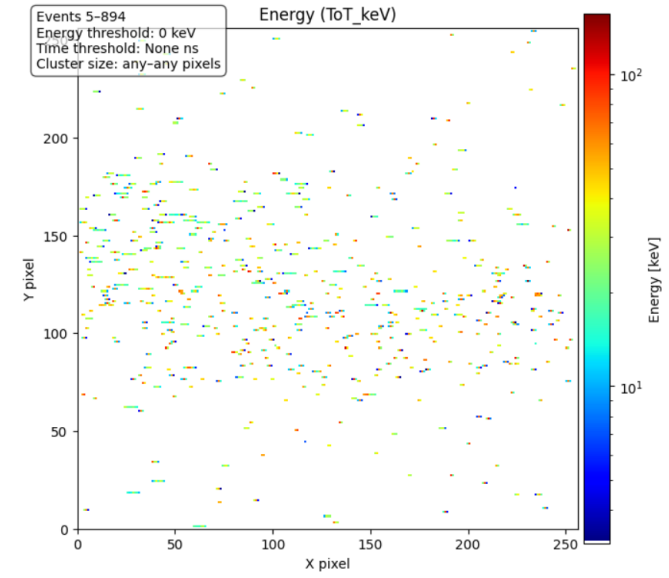
Cluster 1:
y_spread (mean = 0.680, diff = 0.680)
x_spread (mean = -0.476, diff = 0.476)
width (mean = -0.417, diff = 0.417)
Plotting 500 events from cluster 1...



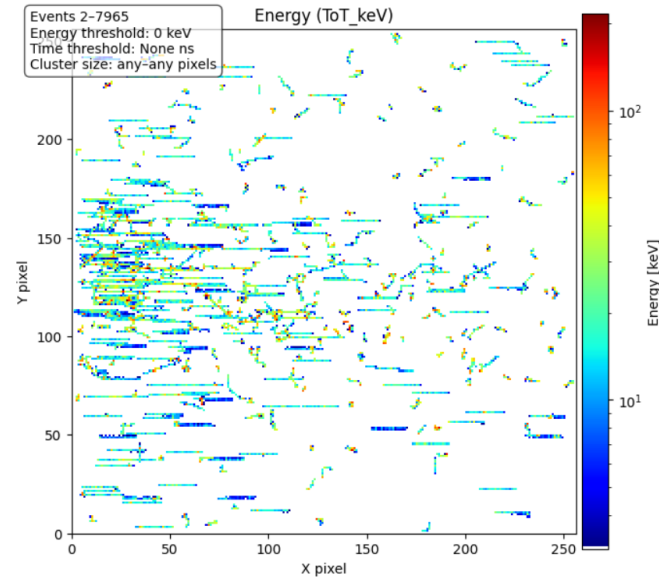
Cluster 2:
aspect (mean = 0.945, diff = 0.945)
mean_energy (mean = -0.651, diff = 0.651)
y_spread (mean = 0.568, diff = 0.568)
Plotting 500 events from cluster 2...



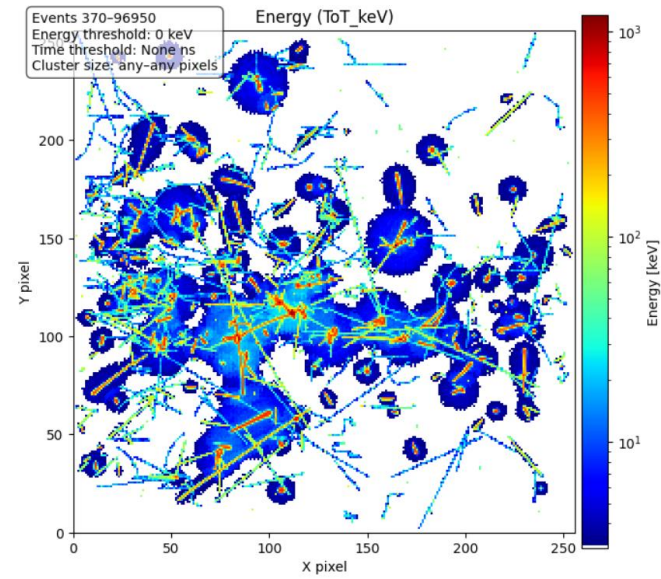
Cluster 3:
y_spread (mean = -0.540, diff = 0.540)
aspect (mean = -0.420, diff = 0.420)
height (mean = -0.370, diff = 0.370)
Plotting 500 events from cluster 3...



Cluster 4:
width (mean = 2.314, diff = 2.314)
x_spread (mean = 2.238, diff = 2.238)
outer_pixels (mean = 2.120, diff = 2.120)
Plotting 500 events from cluster 4...



Cluster 5:
height (mean = 7.165, diff = 7.165)
volcano_fraction (mean = 6.752, diff = 6.752)
y_spread (mean = 5.984, diff = 5.984)
Plotting 500 events from cluster 5...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

PCA

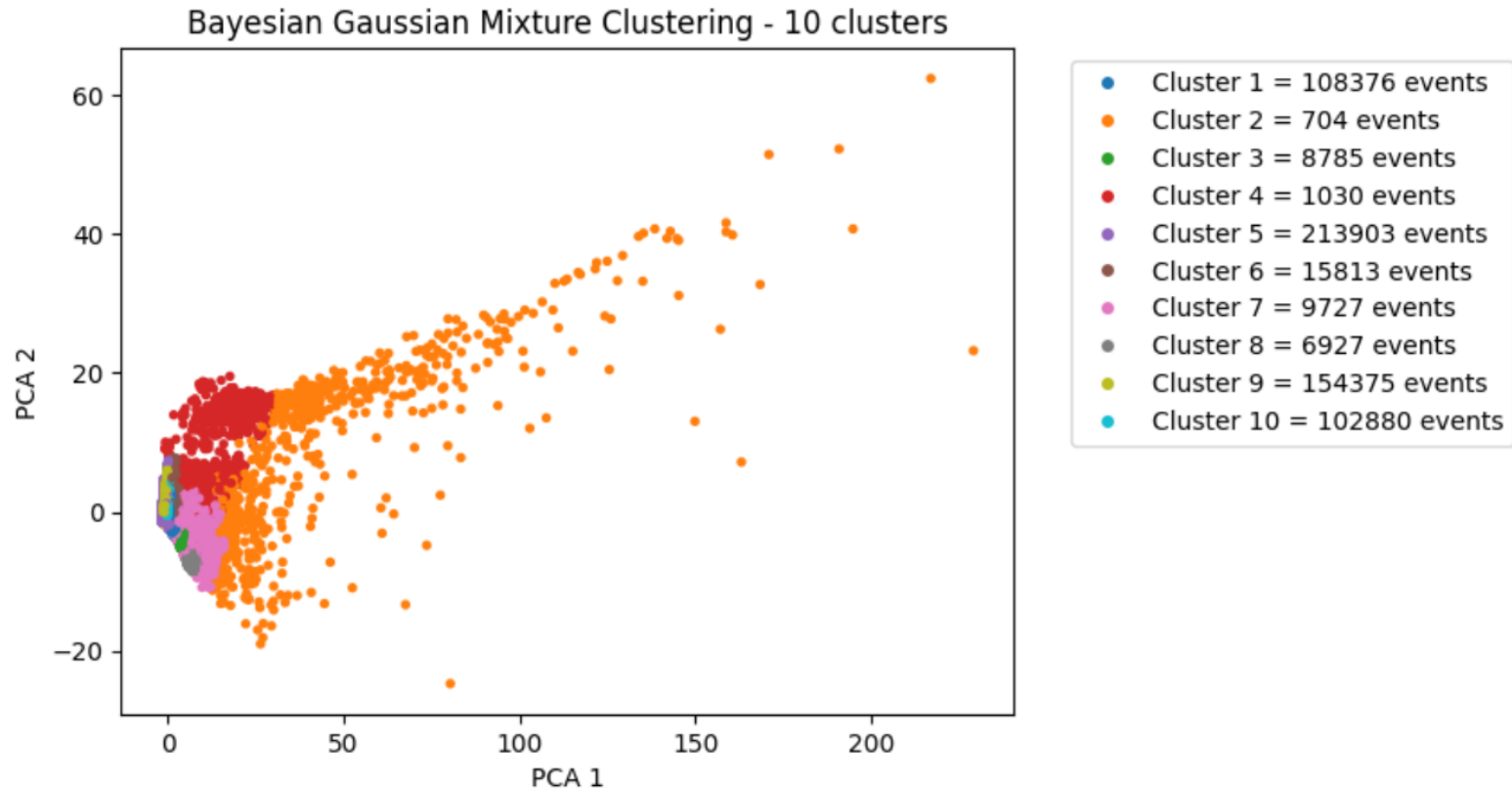
t-SNE

UMAP

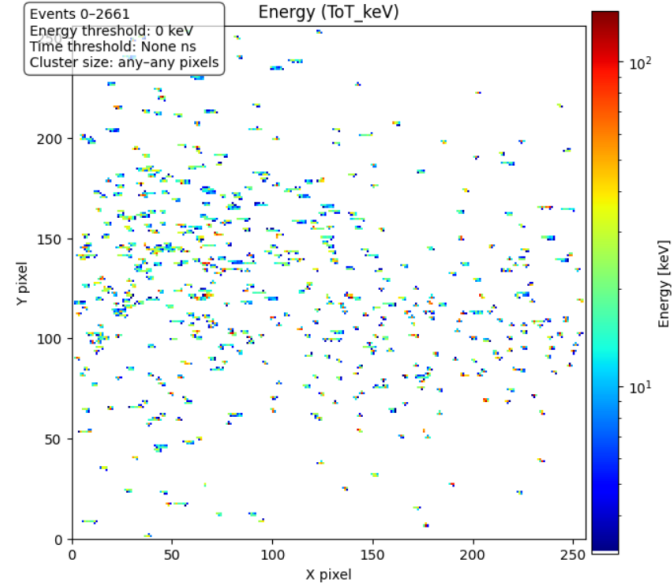
```

bgmm = BayesianGaussianMixture(
    n_components=10, # maximum number of Gaussian clusters
    covariance_type='diag', #'full', 'tied', 'spherical' # each Gaussian has a diagonal covariance matrix
    weight_concentration_prior_type='dirichlet_process', # allows the model to automatically turn off components if they're unnecessary
    weight_concentration_prior=1e-4, # small values push the model to favor fewer clusters
    mean_precision_prior=1.0, # controls how tightly cluster means are pulled toward the global mean of the data
    reg_covar=1e-6, # adds a tiny value to the diagonal of covariance matrices
    max_iter=1000, # maximum number of iterations for the EM (Expectation-Maximization) algorithm
    tol=1e-4, # convergence threshold for EM
    init_params='kmeans', #'random' # how initial cluster means are chosen
    n_init=3, # number of times to run EM with different initializations

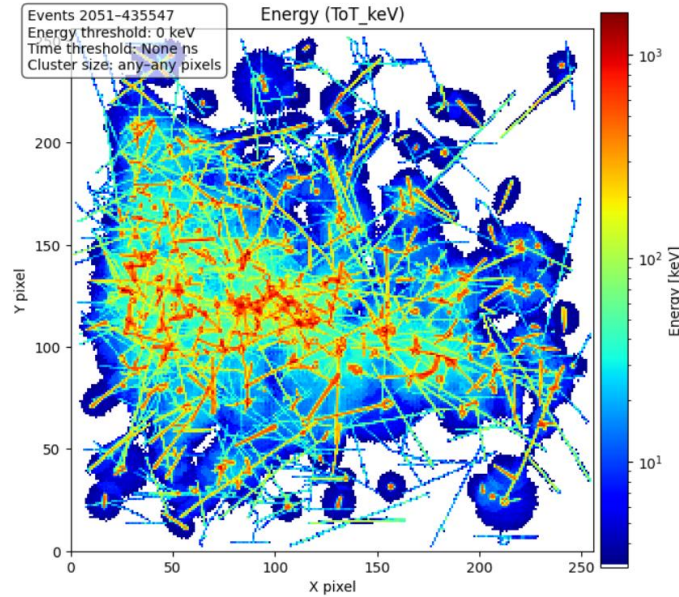
```



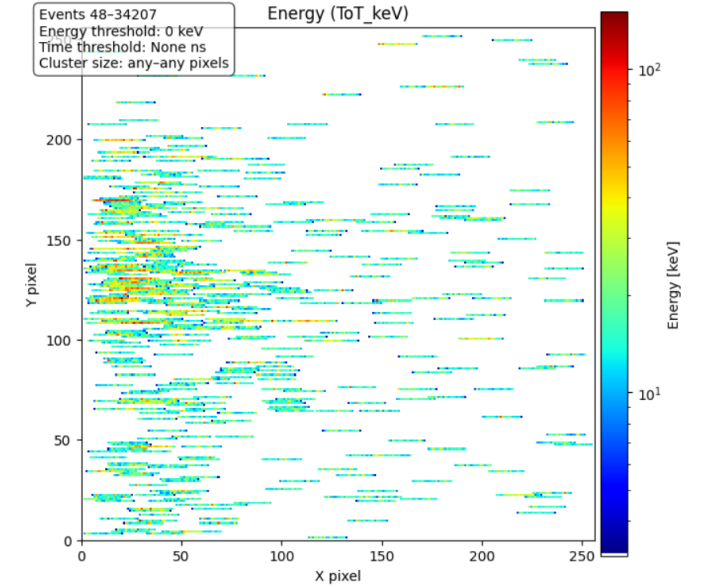
Cluster 1:
 aspect (mean = 0.867, diff = 0.867)
 mean_energy (mean = -0.607, diff = 0.607)
 y_spread (mean = 0.564, diff = 0.564)
 Plotting 500 events from cluster 1...



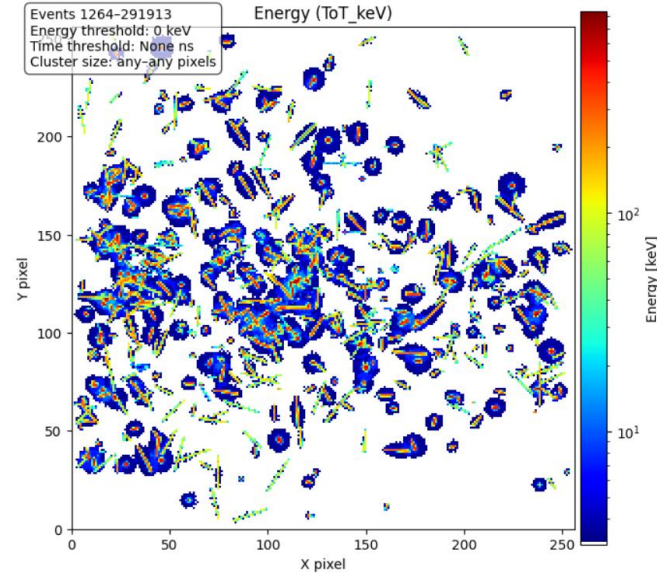
Cluster 2:
 num_hits (mean = 18.634, diff = 18.634)
 total_energy (mean = 18.287, diff = 18.287)
 height (mean = 17.487, diff = 17.487)
 Plotting 500 events from cluster 2...



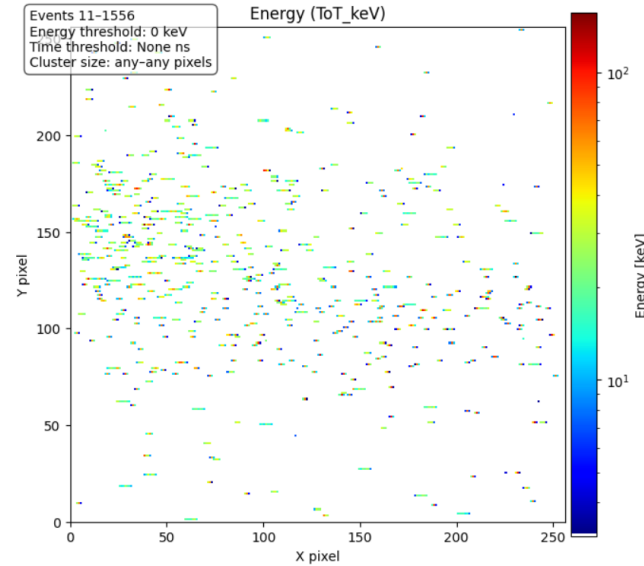
Cluster 3:
 x_spread (mean = 4.848, diff = 4.848)
 width (mean = 4.780, diff = 4.780)
 outer_pixels (mean = 2.980, diff = 2.980)
 Plotting 500 events from cluster 3...



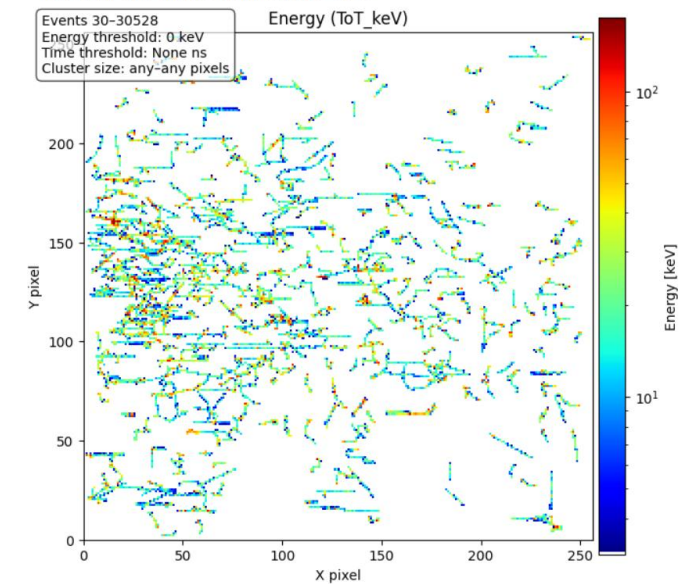
Cluster 4:
 volcano_fraction (mean = 14.112, diff = 14.112)
 max_energy (mean = 8.536, diff = 8.536)
 height (mean = 5.268, diff = 5.268)
 Plotting 500 events from cluster 4...



Cluster 5:
 y_spread (mean = -0.540, diff = 0.540)
 aspect (mean = -0.420, diff = 0.420)
 mean_energy (mean = -0.400, diff = 0.400)
 Plotting 500 events from cluster 5...



Cluster 7:
 height (mean = 3.130, diff = 3.130)
 y_spread (mean = 3.048, diff = 3.048)
 outer_pixels (mean = 2.574, diff = 2.574)
 Plotting 500 events from cluster 7...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

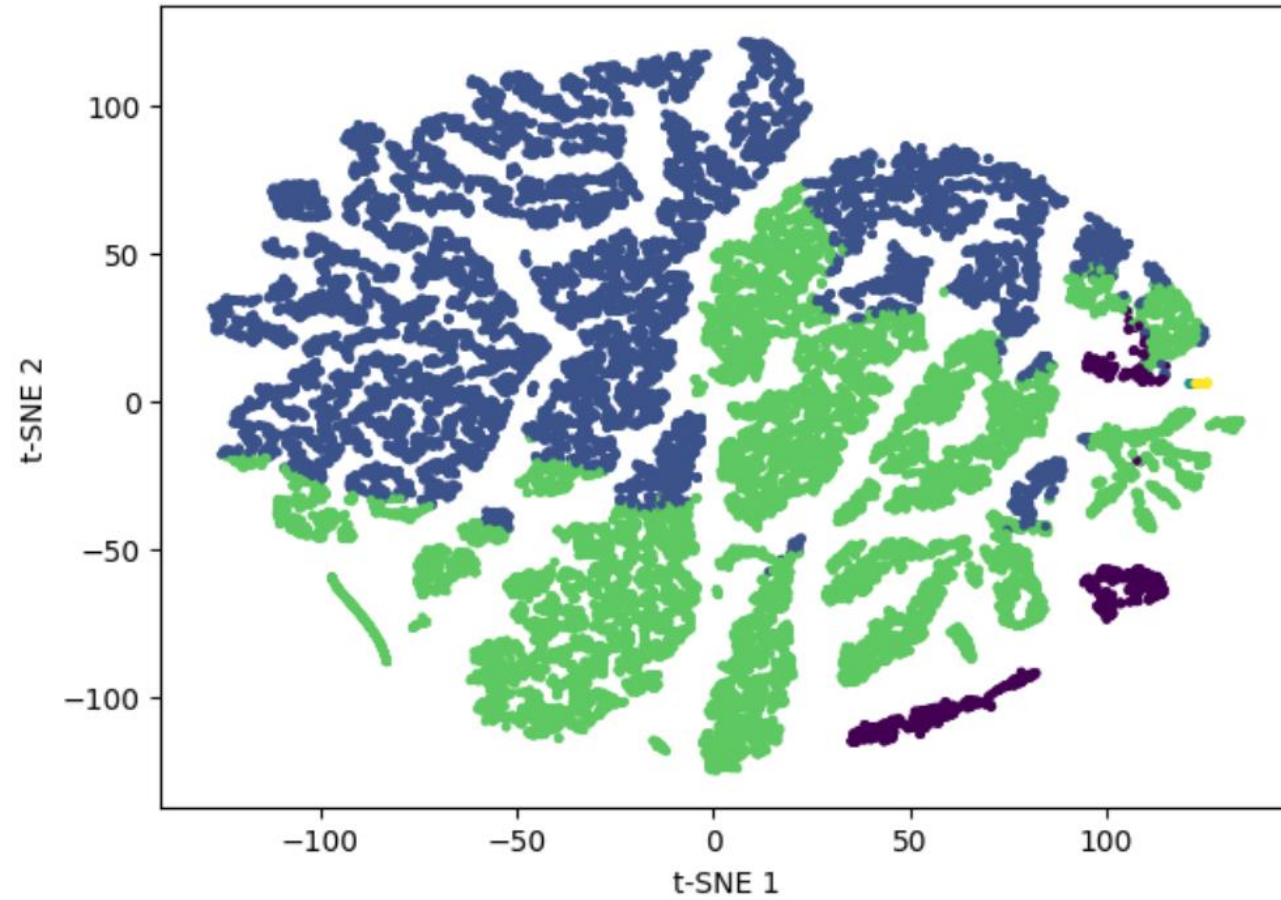
Visualisation

PCA

t-SNE

UMAP

KMeans clustering (t-SNE visualization) (5 clusters)



- Cluster 1 = 1325 events
- Cluster 2 = 19334 events
- Cluster 3 = 12 events
- Cluster 4 = 19290 events
- Cluster 5 = 39 events

Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

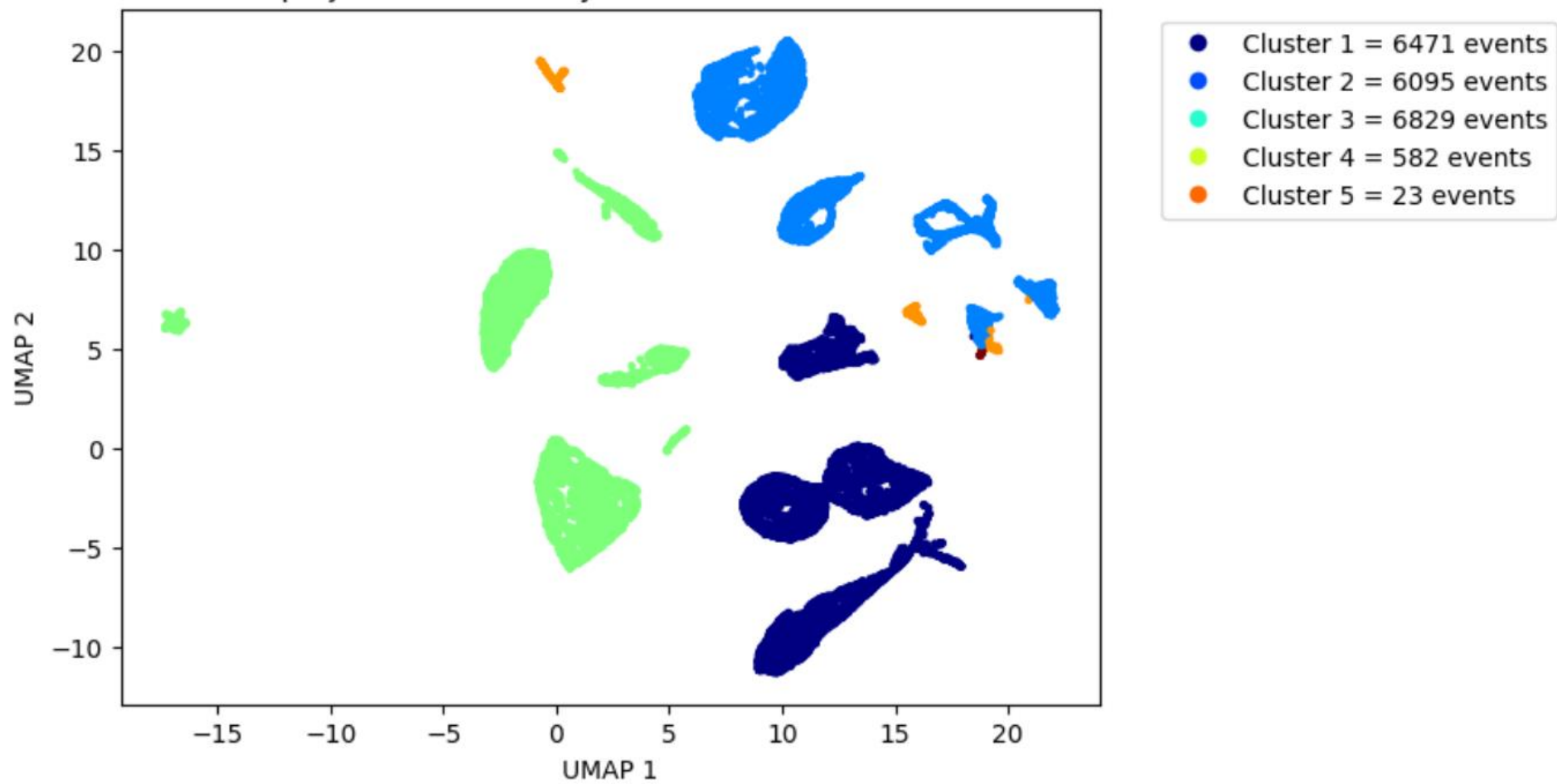
Visualisation

PCA

t-SNE

UMAP

UMAP projection colored by K-Means clusters (5 clusters)



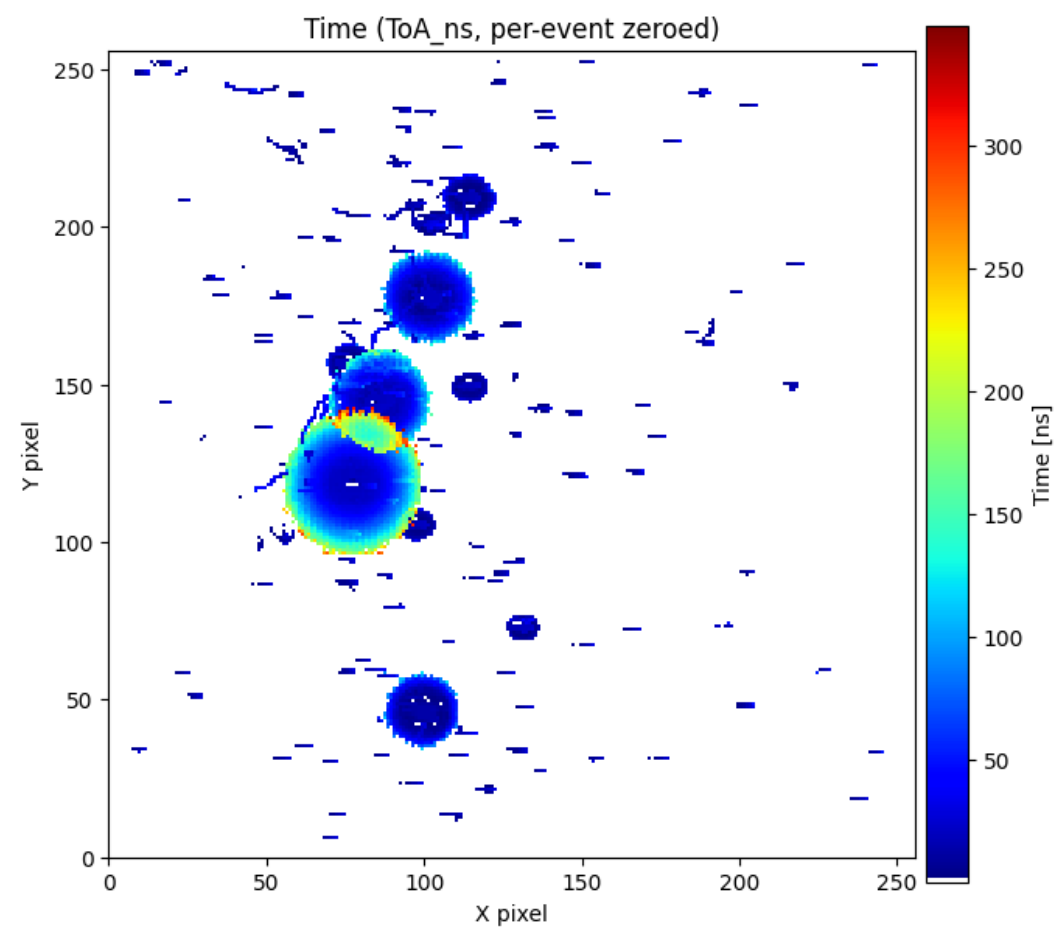
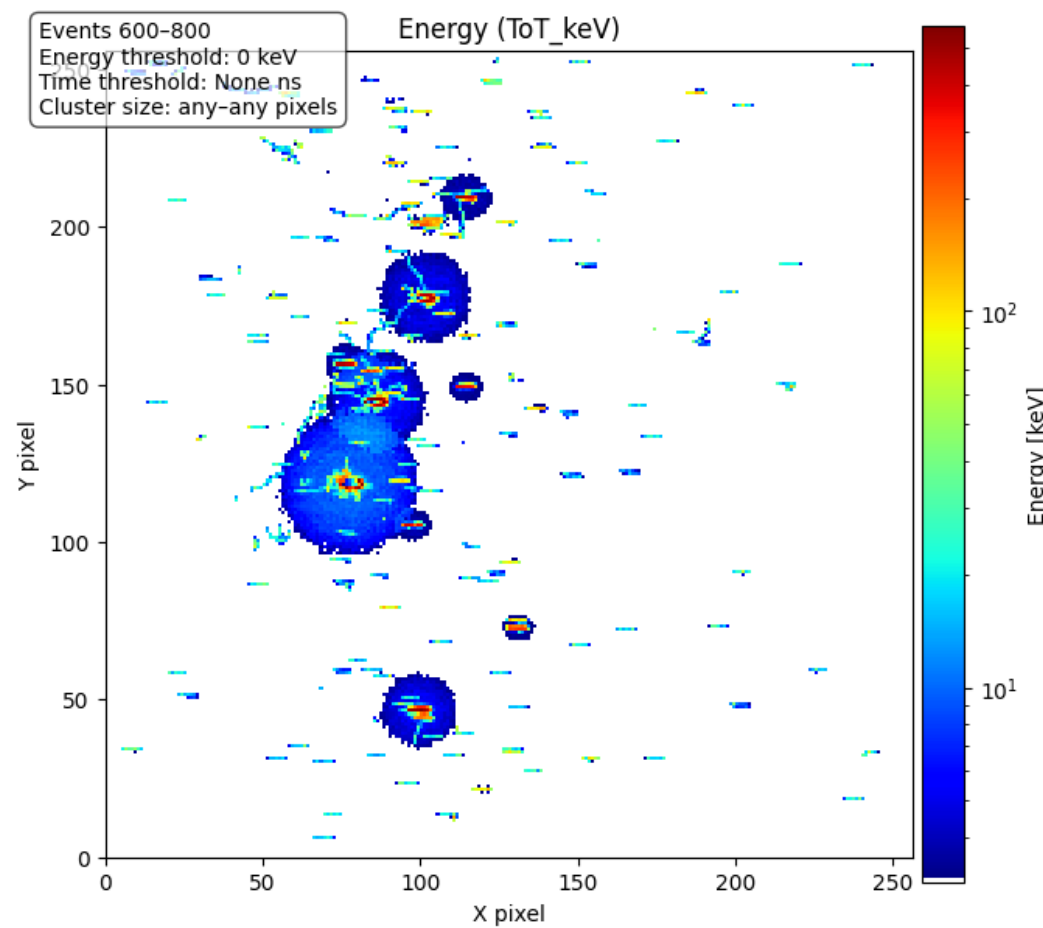
0deg

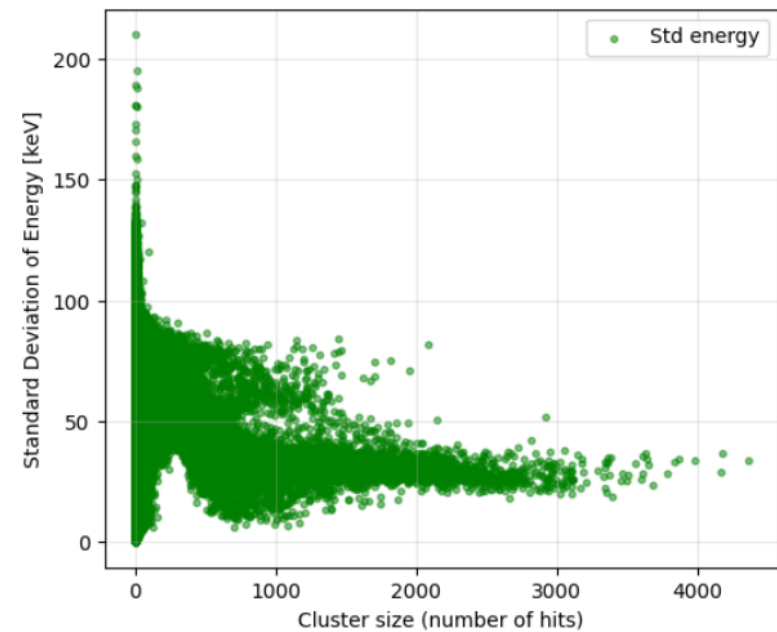
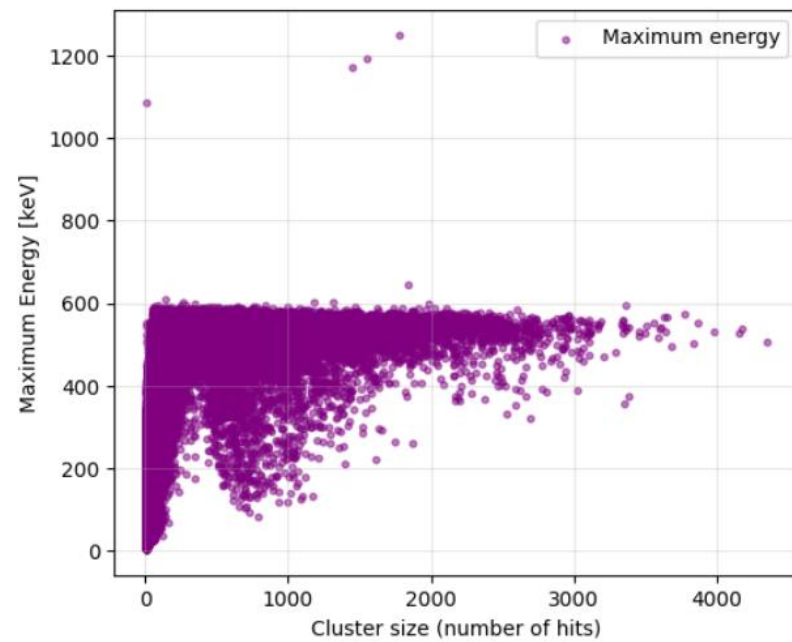
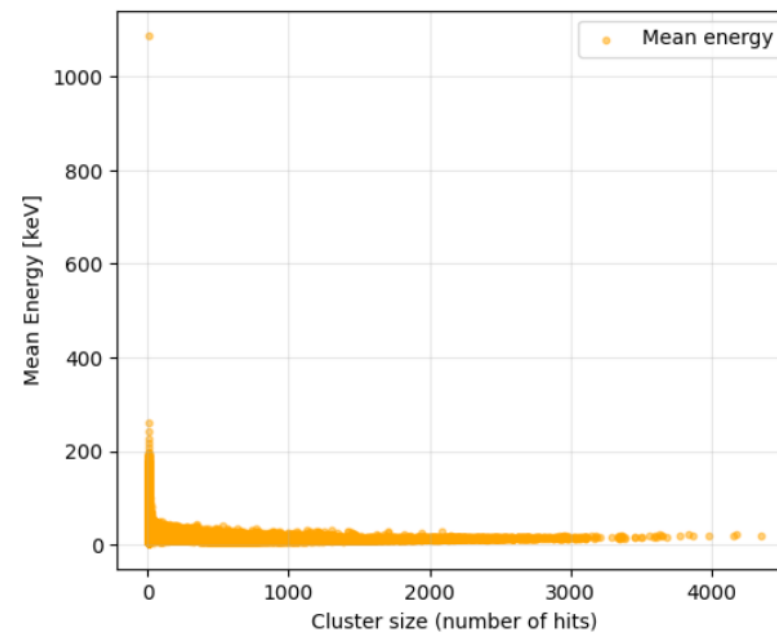
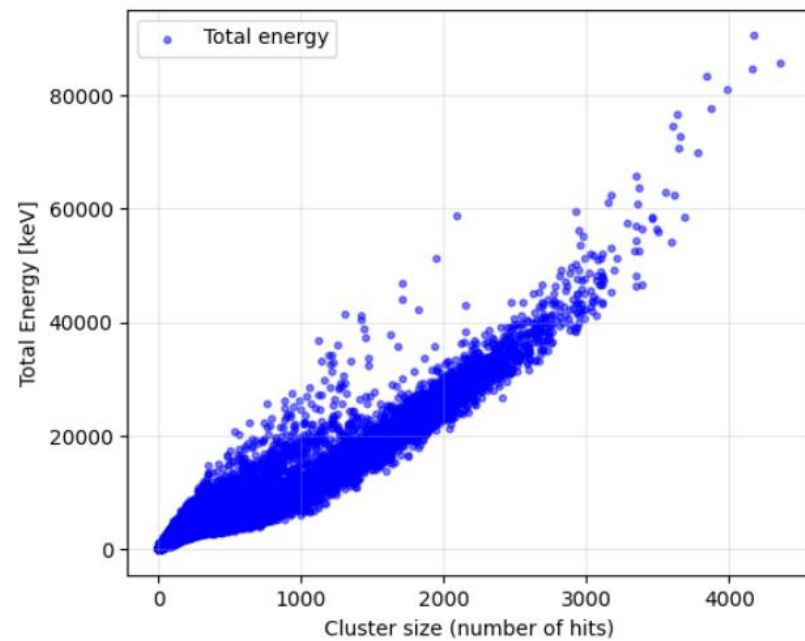
25deg

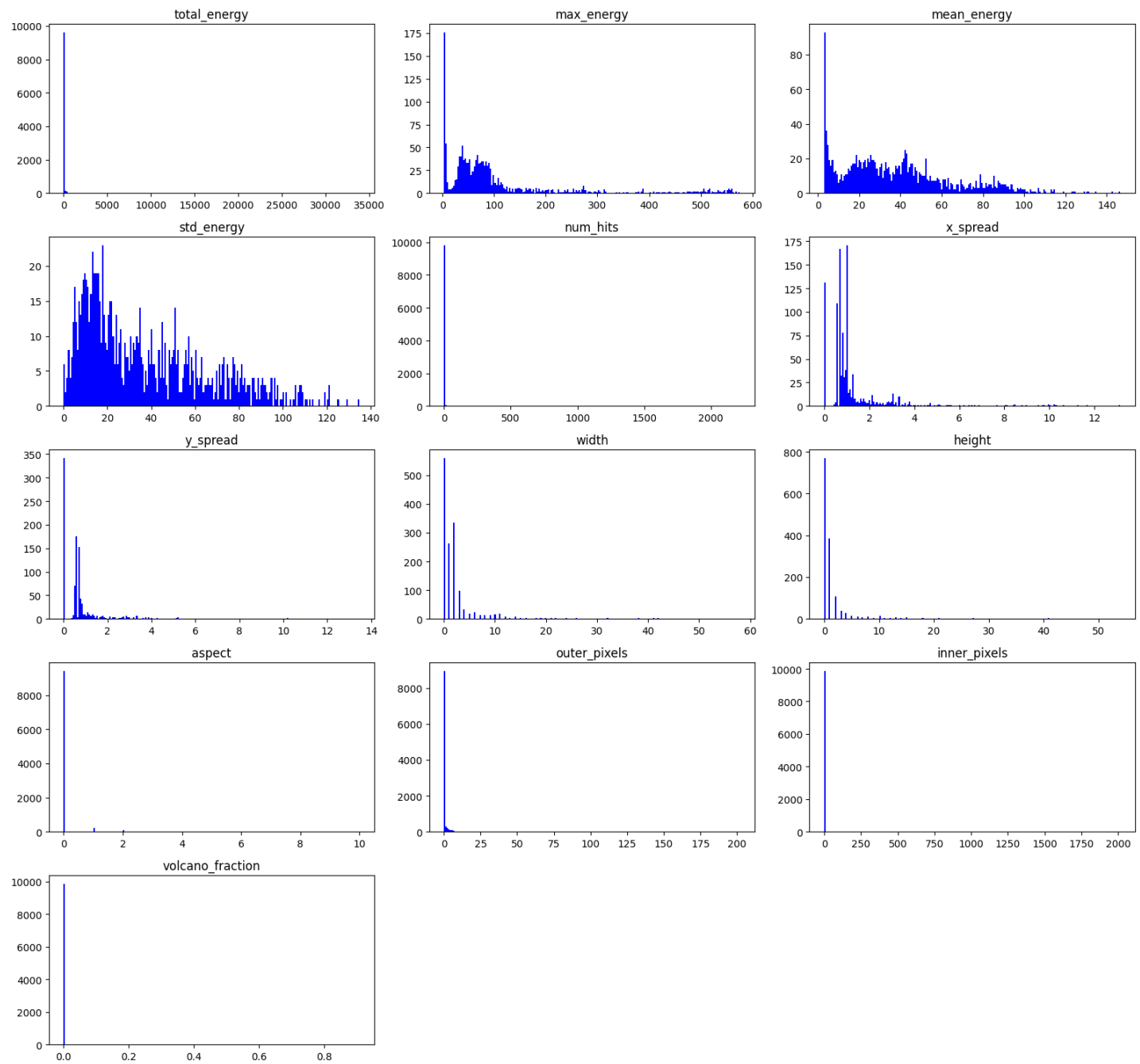
50deg

75deg

Mixed beam







Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

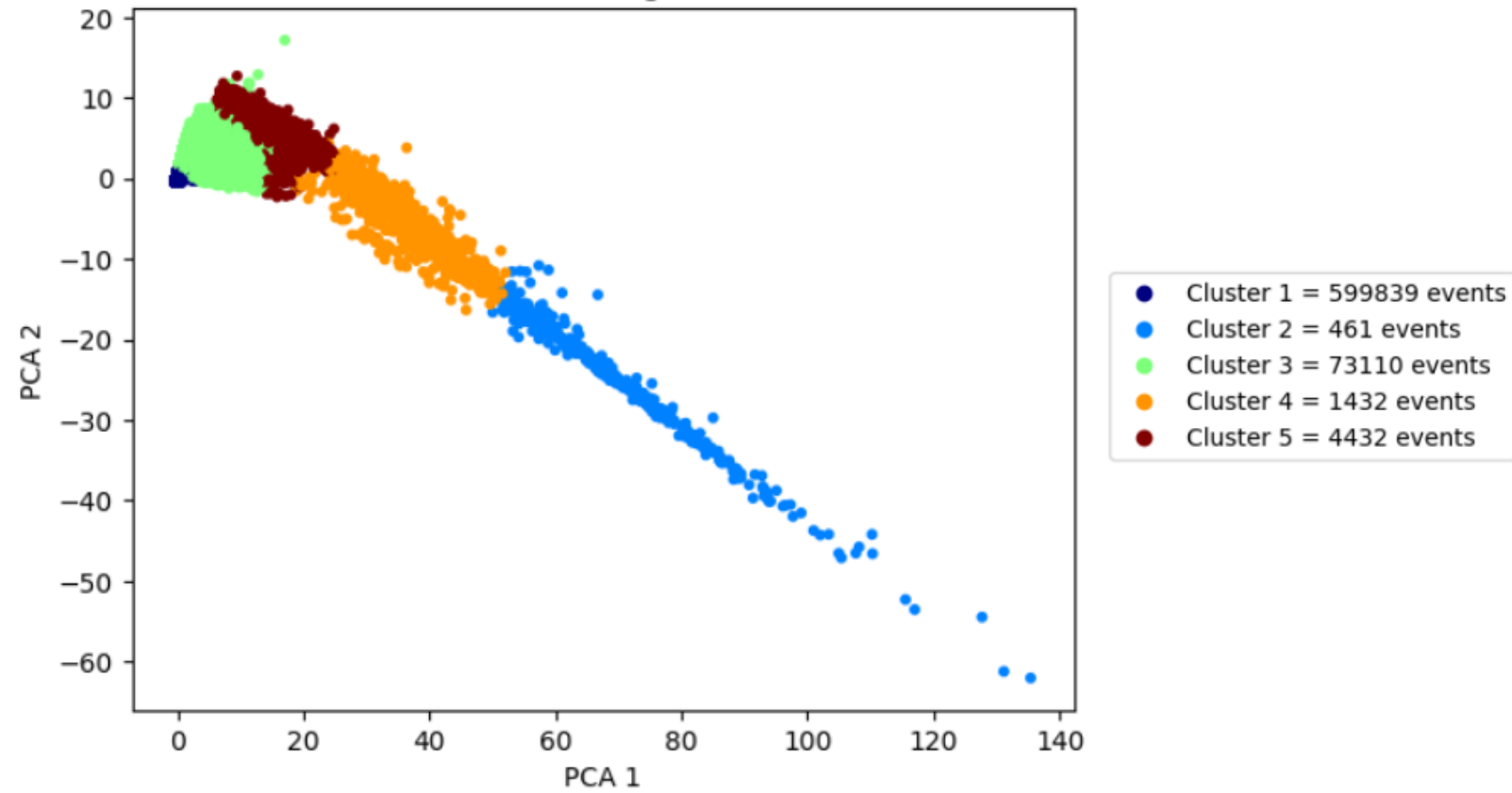
PCA

t-SNE

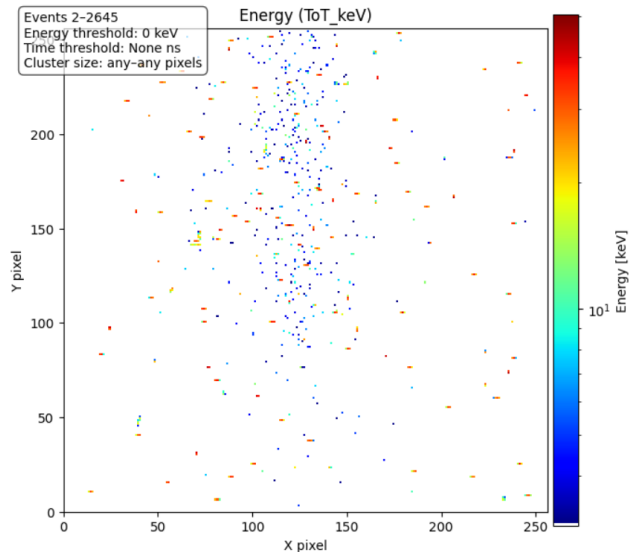
UMAP

Number of clusters for K-Means: 5

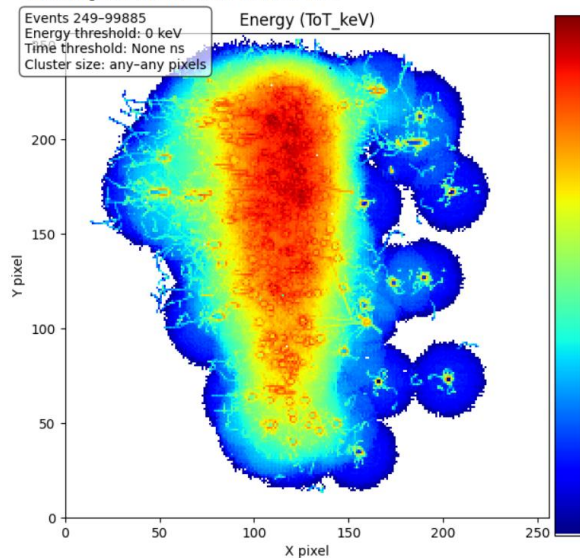
K-means Clustering - 5 clusters



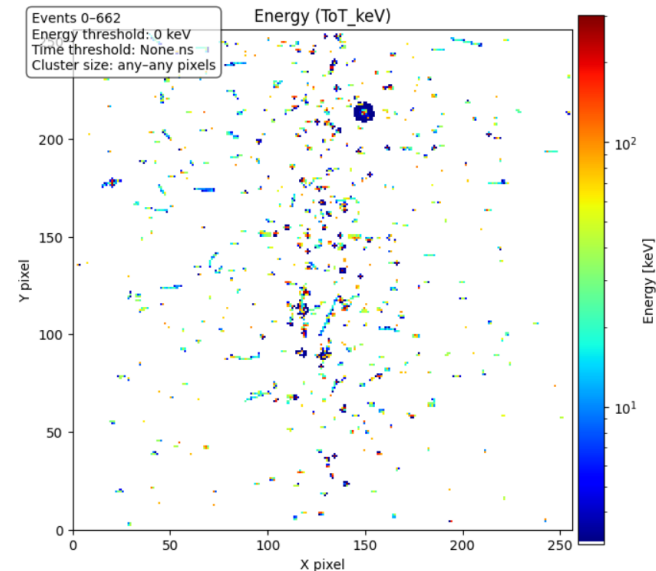
Cluster 1:
 mean_energy (centroid = -0.311, diff = 0.311)
 std_energy (centroid = -0.264, diff = 0.264)
 max_energy (centroid = -0.248, diff = 0.248)
 Plotting 500 events from cluster 1...



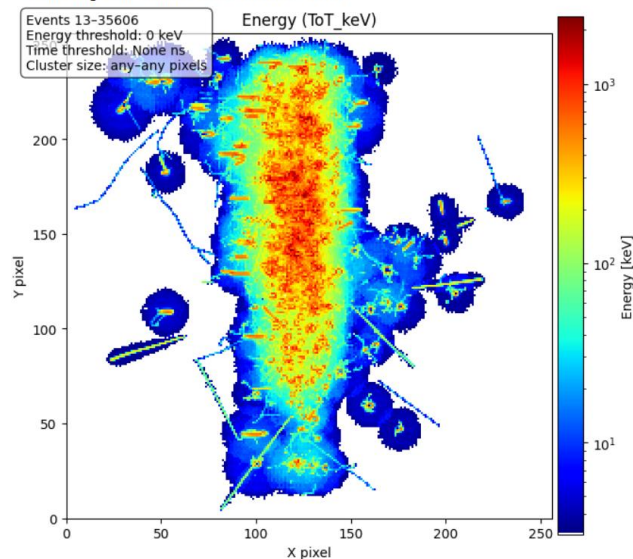
Cluster 2:
 inner_pixels (centroid = 30.972, diff = 30.972)
 num_hits (centroid = 30.479, diff = 30.479)
 total_energy (centroid = 29.964, diff = 29.964)
 Plotting 461 events from cluster 2...



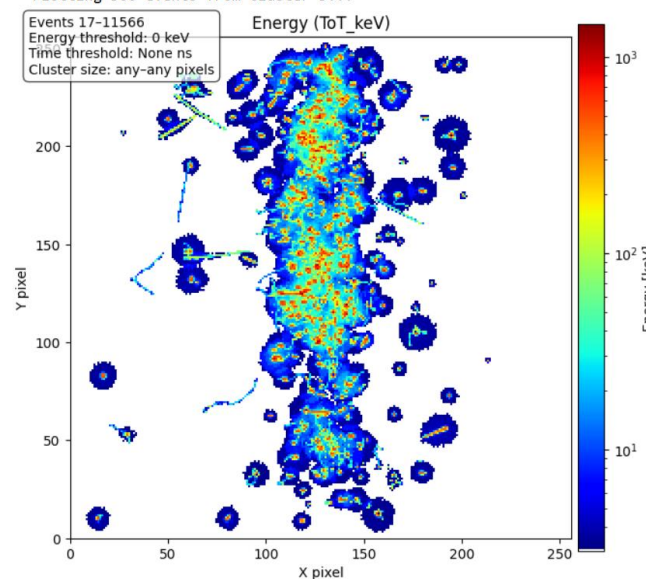
Cluster 3:
 mean_energy (centroid = 2.470, diff = 2.470)
 std_energy (centroid = 1.818, diff = 1.818)
 aspect (centroid = 1.380, diff = 1.380)
 Plotting 500 events from cluster 3...



Cluster 4:
 height (centroid = 13.245, diff = 13.245)
 outer_pixels (centroid = 12.874, diff = 12.874)
 width (centroid = 12.608, diff = 12.608)
 Plotting 500 events from cluster 4...



Cluster 5:
 volcano_fraction (centroid = 11.651, diff = 11.651)
 max_energy (centroid = 7.707, diff = 7.707)
 std_energy (centroid = 4.756, diff = 4.756)
 Plotting 500 events from cluster 5...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

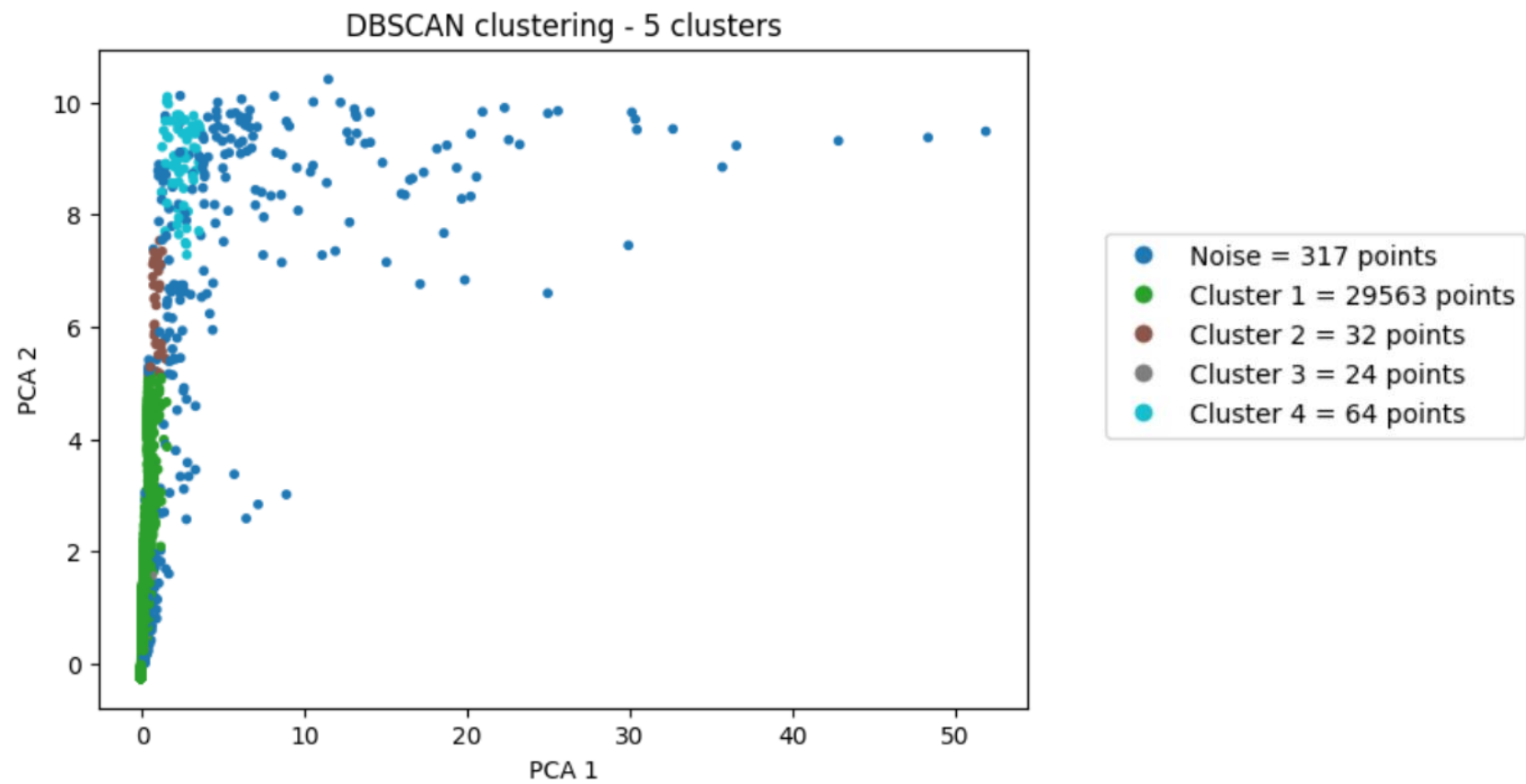
Visualisation

PCA

t-SNE

UMAP

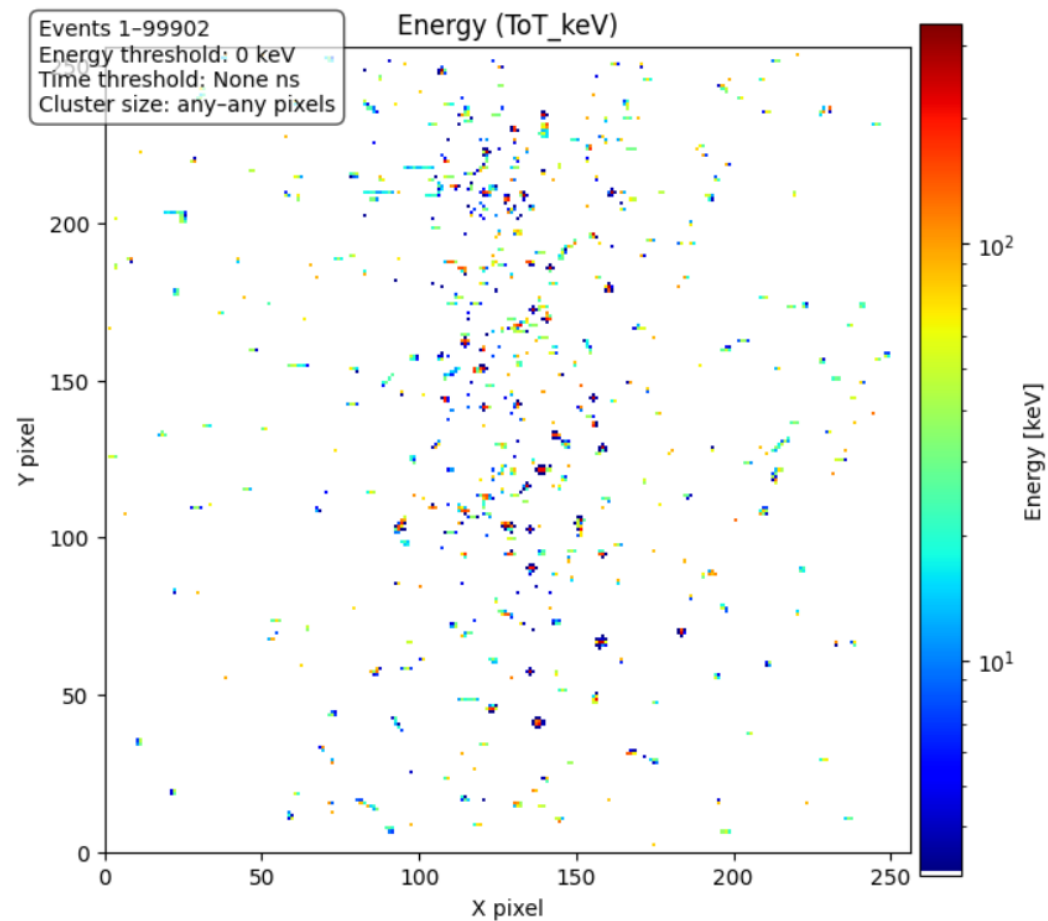
DBSCAN eps (neighborhood radius): 2
DBSCAN min_samples (minimum cluster size): 20



Cluster 1:

x_spread (cluster mean = -0.106, diff = 0.106)
width (cluster mean = -0.104, diff = 0.104)
max_energy (cluster mean = -0.103, diff = 0.103)

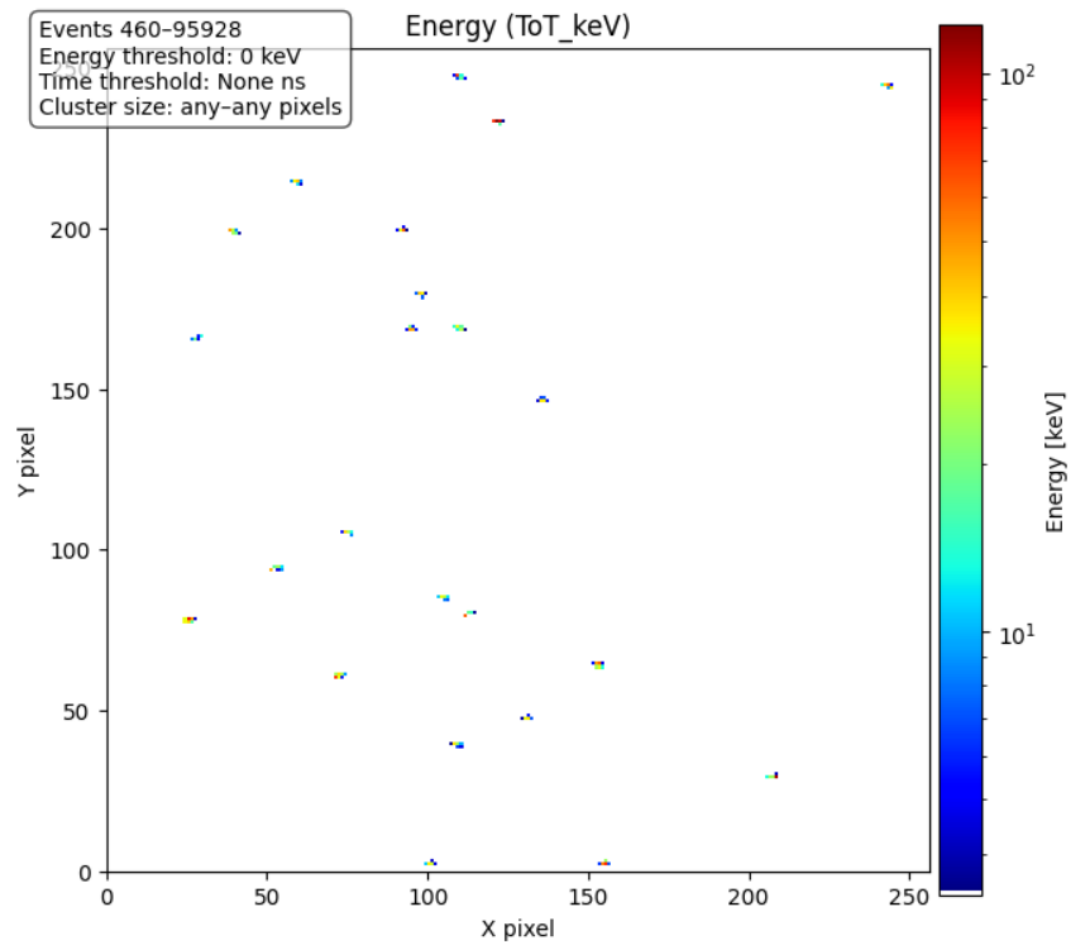
Plotting 500 events from cluster 1...



Cluster 2:

aspect (cluster mean = 6.011, diff = 6.011)
x_spread (cluster mean = 1.583, diff = 1.583)
std_energy (cluster mean = 1.156, diff = 1.156)

Plotting 23 events from cluster 2...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

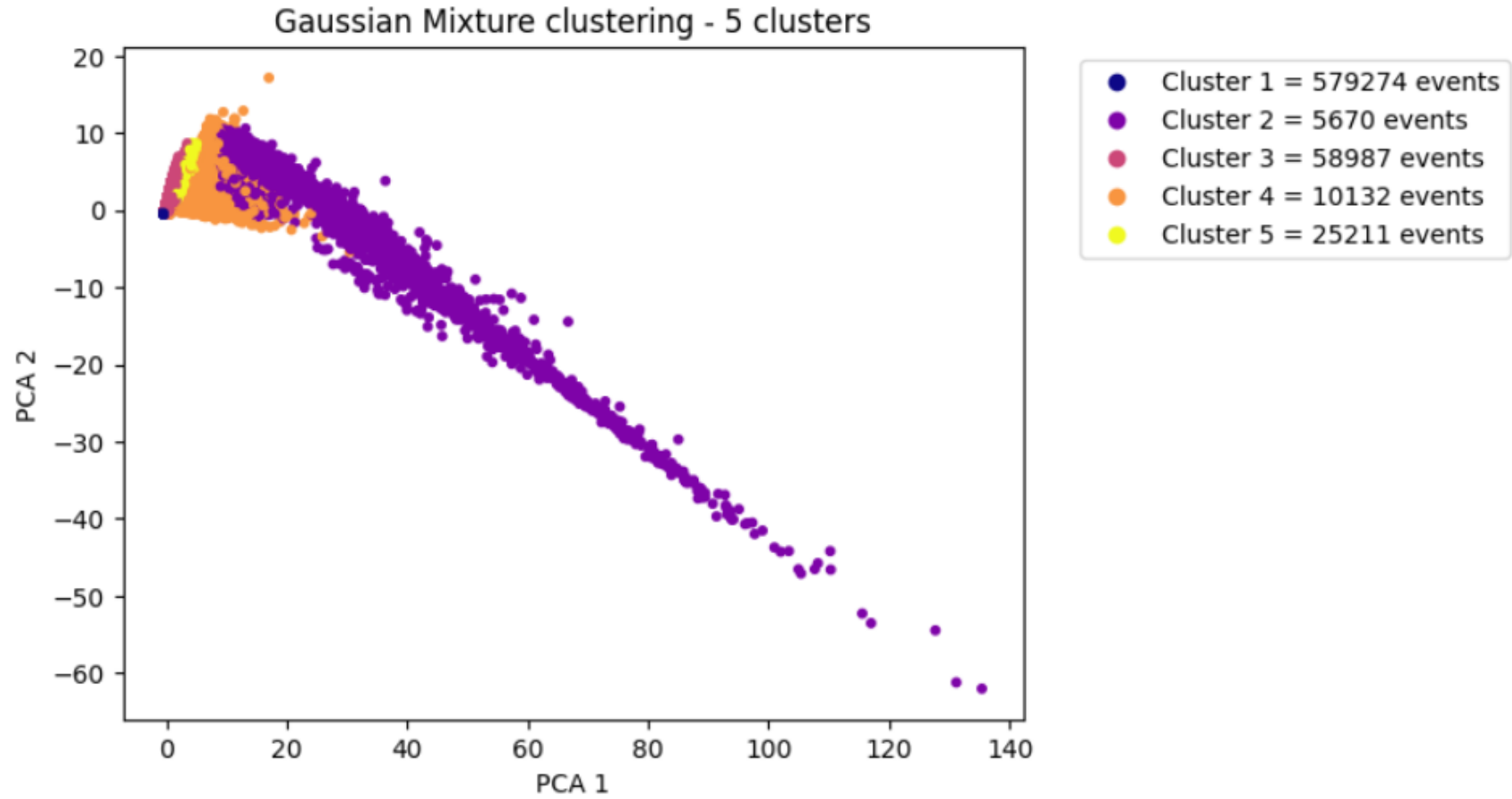
Visualisation

PCA

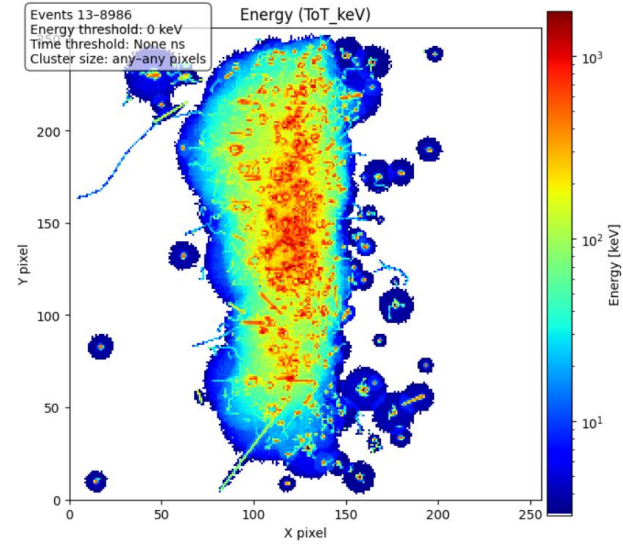
t-SNE

UMAP

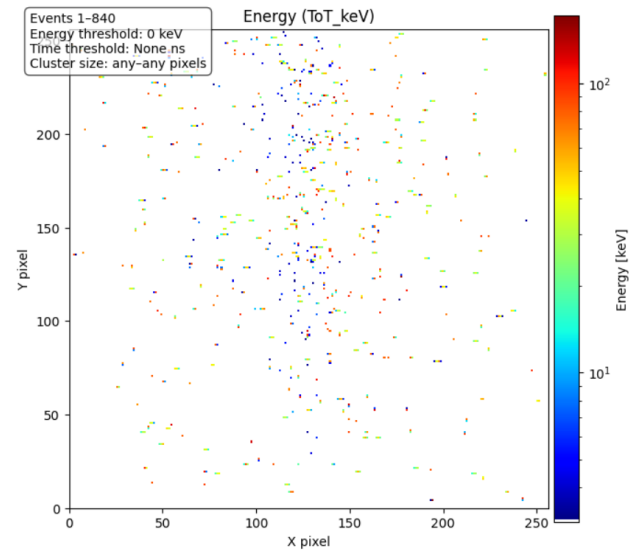
Number of Gaussian clusters: 5



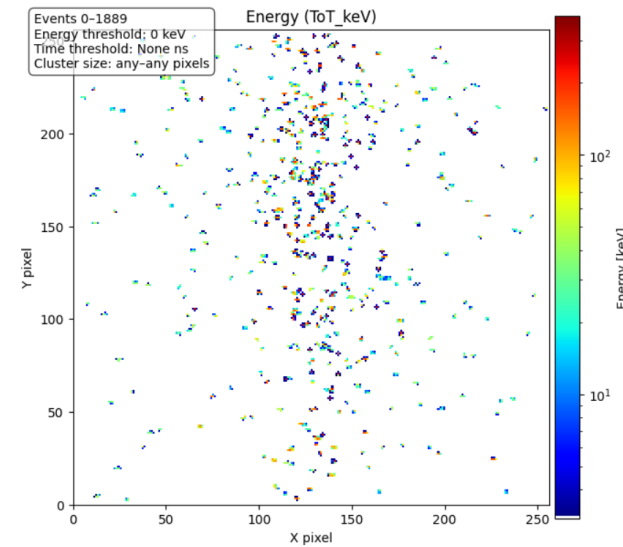
Cluster 2:
 height (mean = 8.645, diff = 8.645)
 y_spread (mean = 8.376, diff = 8.376)
 outer_pixels (mean = 8.372, diff = 8.372)
 Plotting 500 events from cluster 2...



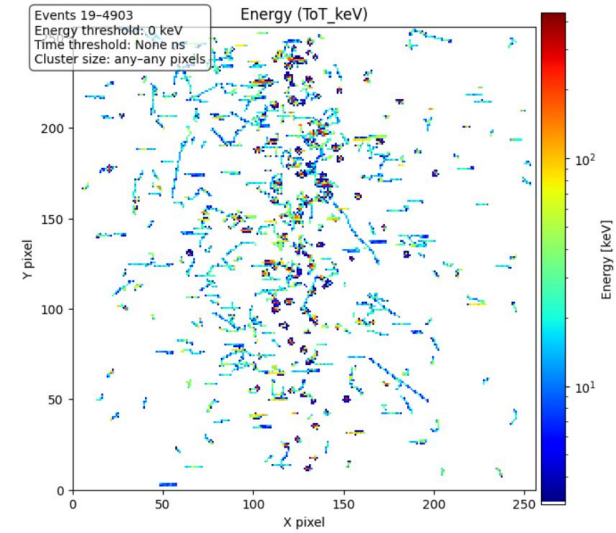
Cluster 3:
 mean_energy (mean = 2.098, diff = 2.098)
 std_energy (mean = 0.769, diff = 0.769)
 max_energy (mean = 0.686, diff = 0.686)
 Plotting 500 events from cluster 3...



Cluster 5:
 aspect (mean = 2.974, diff = 2.974)
 std_energy (mean = 2.703, diff = 2.703)
 mean_energy (mean = 1.938, diff = 1.938)
 Plotting 500 events from cluster 5...



Cluster 4:
 aspect (mean = 3.434, diff = 3.434)
 x_spread (mean = 2.667, diff = 2.667)
 width (mean = 2.136, diff = 2.136)
 Plotting 500 events from cluster 4...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

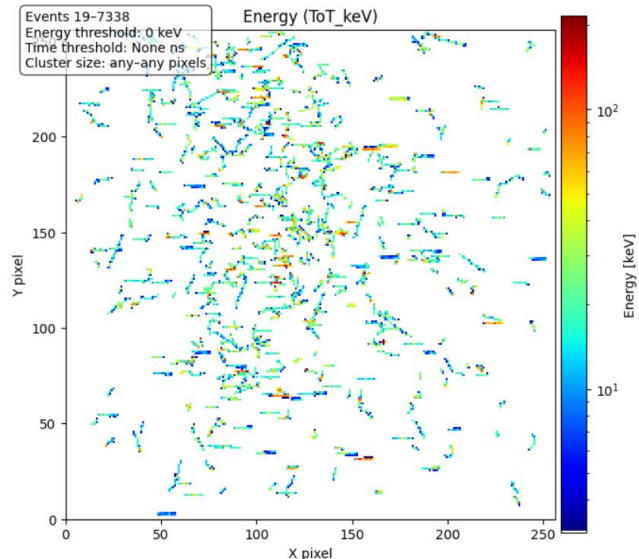
PCA

t-SNE

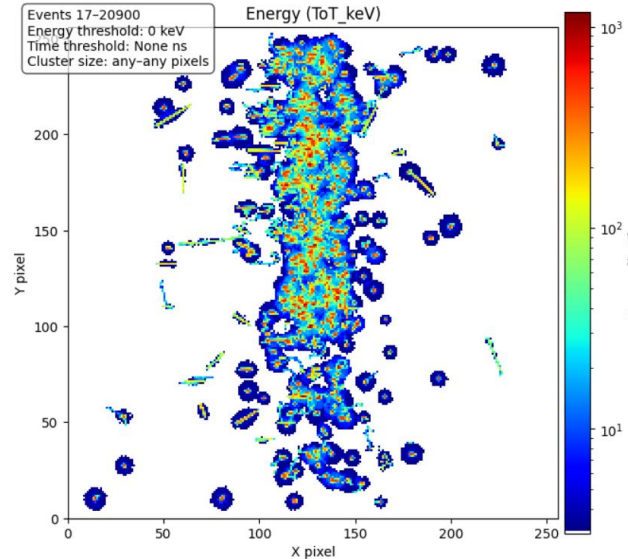
UMAP



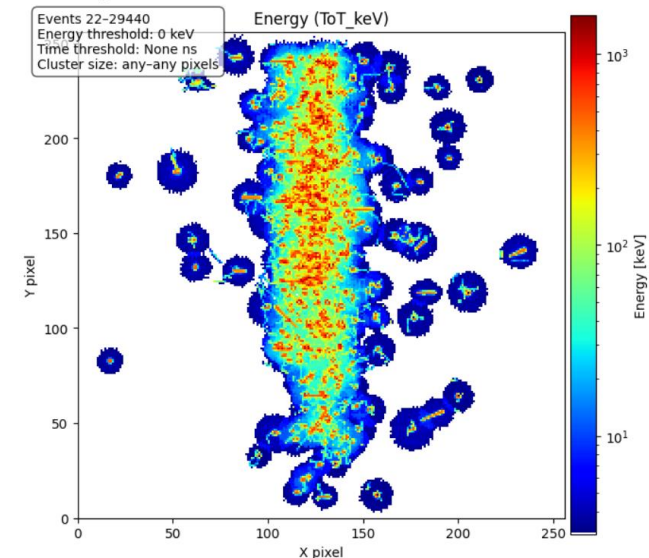
Cluster 10:
 aspect (mean = 4.447, diff = 4.447)
 x_spread (mean = 3.049, diff = 3.049)
 width (mean = 2.429, diff = 2.429)
 Plotting 500 events from cluster 10...



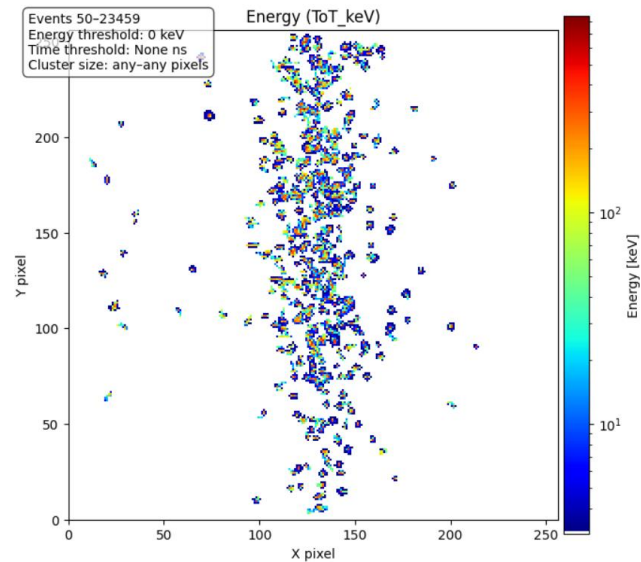
Cluster 2:
 volcano_fraction (mean = 10.068, diff = 10.068)
 max_energy (mean = 7.275, diff = 7.275)
 std_energy (mean = 4.503, diff = 4.503)
 Plotting 500 events from cluster 2...



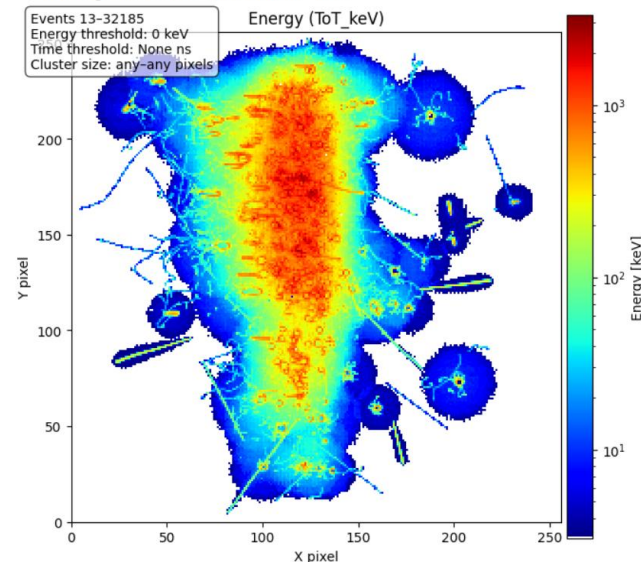
Cluster 3:
 volcano_fraction (mean = 9.155, diff = 9.155)
 max_energy (mean = 8.843, diff = 8.843)
 height (mean = 8.121, diff = 8.121)
 Plotting 500 events from cluster 3...



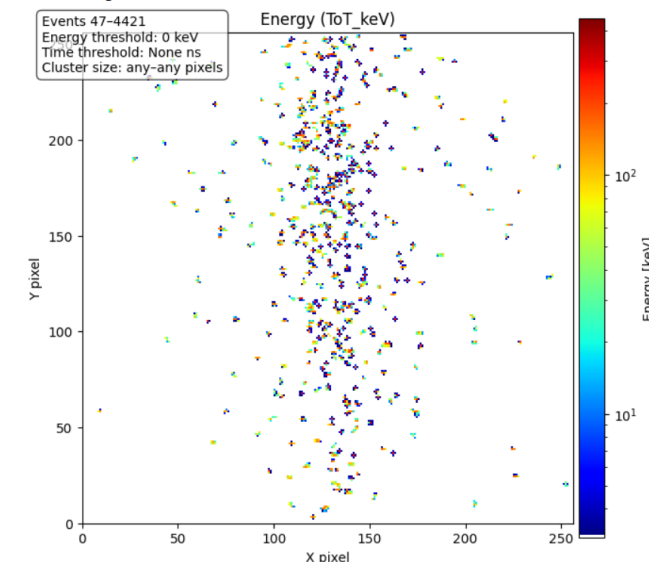
Cluster 4:
 volcano_fraction (mean = 6.442, diff = 6.442)
 std_energy (mean = 5.332, diff = 5.332)
 max_energy (mean = 5.048, diff = 5.048)
 Plotting 500 events from cluster 4...



Cluster 5:
 num_hits (mean = 17.185, diff = 17.185)
 inner_pixels (mean = 17.026, diff = 17.026)
 outer_pixels (mean = 16.870, diff = 16.870)
 Plotting 500 events from cluster 5...



Cluster 6:
 std_energy (mean = 4.756, diff = 4.756)
 max_energy (mean = 3.089, diff = 3.089)
 mean_energy (mean = 2.961, diff = 2.961)
 Plotting 500 events from cluster 6...



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

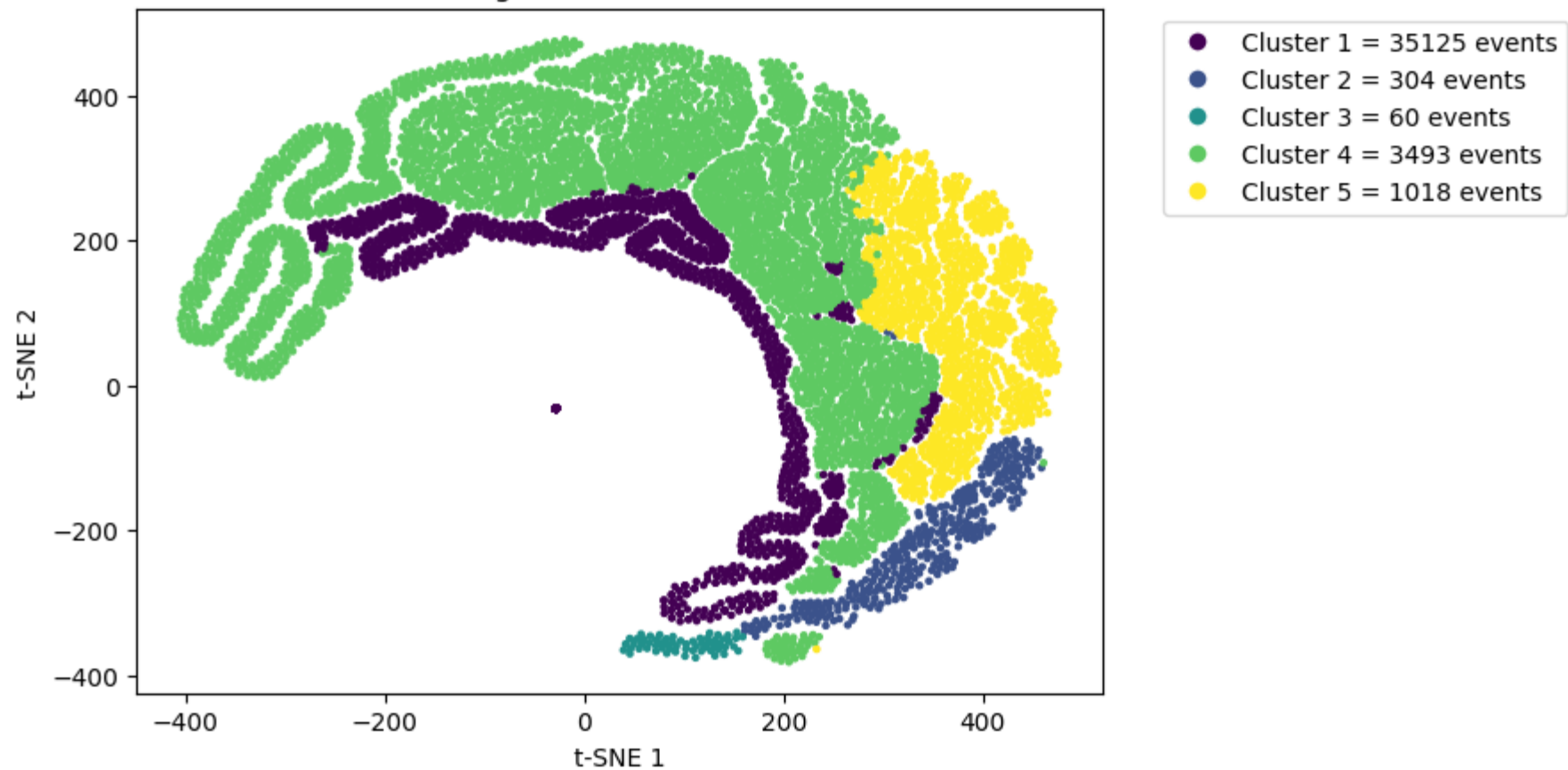
Visualisation

PCA

t-SNE

UMAP

KMeans clustering (t-SNE visualization) (5 clusters)



Clustering

K-means

DBSCAN

Gaussian Mixture

Bayesian Gaussian
Mixture

Visualisation

PCA

t-SNE

UMAP

UMAP projection colored by K-Means clusters (5 clusters)

